



ScienceDirect

Journal of Materials Processing Technology

Date: November 2025

[Show more](#) ✓Research article Full text access [Get rights and content](#) ↗

Research article

Surface microstructuring strategy for suppressing hump defects in ultra-high-power laser penetration welding of thick plates: Experimental and numerical insights

Yiyang Hu ^a, Fei Yan ^b, Zehui Liu ^a, Zhongshun Zhao ^a, Chunming Wang ^a   [Show more](#) ✓

View PDF

[Download full issue](#)

Cite

[Add to Mendeley](#)[Share](#)[10.1016/j.jmatprotec.2025.119075](https://doi.org/10.1016/j.jmatprotec.2025.119075) ↗

Article

Recommended articles

Metrics

Outline

[Highlights](#)[Abstract](#)[Graphical Abstract](#)[Keywords](#)[1. Introduction](#)[2. Experiment and numerical model](#)[3. Mathematical models](#)

Highlights

- Proposed a butt-joint microstructuring approach to suppress hump defects.
- Achieved single-pass welding formation of 20 mm thick plates.
- Developed a high-fidelity numerical model incorporating microstructural morphology.
- Revealed the mechanism of hump defects suppression by surface microstructure.

Abstract

Ultra-high-power laser enables single-pass welding of thick-section components with high efficiency and low cost. However, its application is hindered by the frequent occurrence of hump defects. In this work, a method based on surface microstructuring of the butt joint cross section is proposed to suppress hump defects, achieving well-formed single-pass welding of 20 mm thick stainless steel. A computational fluid dynamics model incorporating surface microstructural features is developed and shows good agreement with experimental results. Surface microstructuring introduces two direct effects: it facilitates molten pool expansion and enhances laser energy absorption. These effects promote the transition of the keyhole from a non-penetrating to a penetrating state. The melt near the keyhole wall exhibits a higher tangential velocity while the downward flow of molten material is suppressed. No low-speed zone is observed at the bottom of the molten pool, enabling rear-side melt contraction driven by surface tension, which contributes to hump suppression. The resulting weld exhibits finer grains and a higher fraction of low-angle grain boundaries. The tensile strength and elongation reach 96% and 65% of those of the base metal, respectively. The changes in weld microstructure are attributed to the higher temperature thermal history experienced by the molten pool. The improved mechanical properties are primarily due to the enhanced weld formation quality. The proposed method requires no additional auxiliary equipment during welding and provides a promising solution for the industrial application of ultra-high-power laser single-pass welding in thick-section component manufacturing.

Graphical Abstract

4. Results

5. Discussions

6. Conclusions

CRediT authorship contribution statement

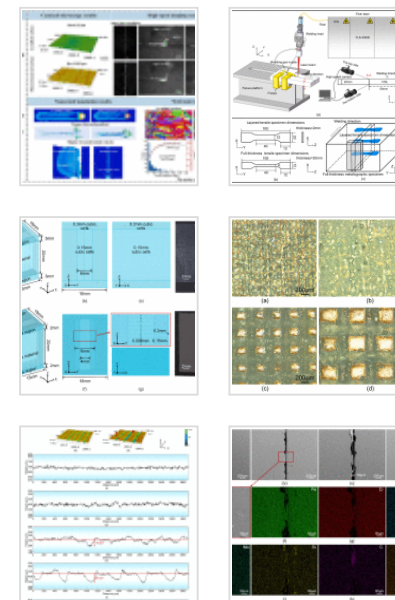
Declaration of Competing Interest

Acknowledgments

References

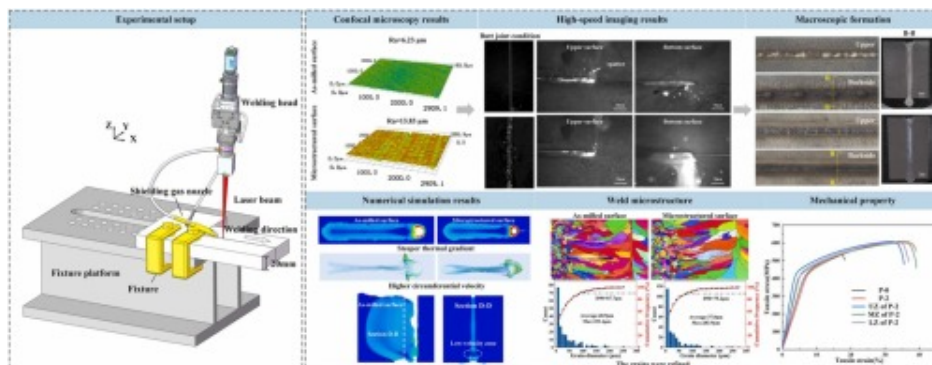
[Show full outline](#) ▼

Figures (20)



[Show 14 more figures](#) ▼

Tables (4)



Download: [Download high-res image \(360KB\)](#)

Download: [Download full-size image](#)

Keywords

Ultra-high power laser penetration welding; Surface microstructuring; Hump defects suppression; Numerical simulation; Marangoni flow modulation

[< Previous article in this issue](#)

[Next article in this issue >](#)

1. Introduction

Thick-walled components possess excellent load-bearing capacity and structural stability, making them critical structural elements in the manufacturing of high-end equipment such as large pressure vessels [1], [2] and nuclear reactor shells [3], [4]. Thick-plate welding is a key manufacturing process for ensuring the strength and safety of large-scale engineering structures [5], [6]. Laser welding employs a high energy density laser beam as the heat source and enables single-pass welding of centimeter-thick materials [7], offering a significant efficiency advantage over conventional arc welding [8]. In recent years, with the decreasing manufacturing cost and continuous power scaling of high

- ▣ Table 1. Chemical compositions of 316L.
- ▣ Table 2. Process parameters for surface ...
- ▣ Table 3. Thermal physical properties of 3...
- ▣ Table 4. Comparison of weld seam width...

power lasers [9], the use of ultra-high-power lasers for single-pass welding of thick-section components has become increasingly feasible for commercial applications. However, the welding process is prone to forming defects such as hump formation [10], which deteriorate the mechanical properties of the joint [11] and hinder the application of this technology.

The direct cause of hump defects is the excessive accumulation of molten material at the bottom, which leads to humps formation after solidification [12]. Existing hump defects suppression methods mainly focus on two aspects: preventing melt accumulation and promoting melt backflow into the molten pool. Katayama [13] and Youhei [14] demonstrated that vacuum or low-pressure environments increased the penetration depth in laser keyhole welding. Arata [15] and Luo [16] conducted welding experiments using a CO₂ laser and a fiber laser, respectively, and both found that vacuum conditions can almost completely suppress the formation of laser-induced plasma. Jiang [17] successfully achieved sound formation of 20 mm thick high-strength steel joints under a reduced ambient pressure environment and found that the molten pool exhibited a lower peak temperature under such conditions. Welding in a vacuum environment helps reduce the molten pool temperature and the recoil pressure caused by metal vaporization, thereby preventing melt accumulation at the bottom of the molten pool. Bachmann [18] and Avilov [19] applied alternating magnetic fields during the welding of aluminum alloys and duplex steels, respectively. The Lorentz force generated provided upward support to the molten pool, thereby suppressing hump defects by promoting melt backflow into the pool. Xu [20] introduced a shielding gas chamber on the backside of the aluminum alloy specimen to prevent the formation of an oxide film on the molten surface. This allowed the surface tension of the molten pool itself to promote melt backflow, effectively avoiding the formation of hump defects. Hu et al. [21] installed a trailing shielding gas nozzle on the backside of the specimen, where the pressure from the gas flow supported the molten pool. Under the influence of the gas flow, the keyhole transitioned from a non-penetrating to a penetrating state, releasing internal pressure and reducing melt accumulation at the keyhole bottom. The above studies, which

employed vacuum conditions, electromagnetic assistance, and gas-flow assistance, have all proven effective in suppressing hump defects during laser penetration welding. However, the use of additional auxiliary equipment increases manufacturing costs and reduces process flexibility, highlighting the need for more adaptable defects suppression strategies.

Thick-plate welding involves a larger butt joint cross section, and the surface condition of the joint cross section has a non-negligible effect on the welding process. Bergström [22] measured the absorptivity of metals to laser irradiation and found that metals with a certain degree of surface roughness exhibited higher absorptivity than polished metals. This increase in absorptivity was attributed to the presence of surface oxides and surface roughness. For stainless steel, when the surface roughness exceeds 1.5 μm , absorptivity increased with increasing roughness. Baumann [23] found that the absorptivity of copper plates to laser irradiation increased twofold with increasing surface roughness. Sokolov pointed out that in thick-section laser beam welding at high power levels (10 kW), the absorptivity is highly dependent on the edge surface roughness, and increasing the roughness can enhance both the weld penetration depth [24] and welding efficiency [25]. Li [26] prepared four levels of surface roughness on Invar 36 alloy using pneumatic shot peening. The welding results showed that rougher surfaces enhanced laser energy absorption, but also led to pore formation and coarser grain structures in the weld. Cai [27] also noted that the butt joint gap significantly affects the formation of hump defects. These studies suggest that introducing a microstructured surface with controlled roughness at the joint cross section can increase laser absorptivity and alter molten pool flow behavior. Cross-sectional microstructuring may therefore serve as a potential strategy for suppressing hump defects in thick-plate welding.

When using laser processing, selecting a grid-shaped scanning strategy is one of the most effective and controllable methods for significantly increasing surface roughness (R_a). Ozan [28] pointed out that cross-hatching remains the most efficient pathway to maximize surface roughness while maintaining dimensional accuracy. Wang [29] demonstrated that the grid-shaped scanning strategy increased the R_a value from 0.12 μm

to 2.1 μm . Alsaigh [30] emphasized that, compared with simpler patterns, the grid-shaped scanning strategy consistently provided an 80–120% improvement in surface roughness. Related studies have also investigated the influence of laser processing parameters on surface roughness. Zhou [31] pointed out that laser power and scanning speed are the key factors affecting surface texture through an extensive literature review. Li [32] investigated the effects of laser power, scanning speed, and pulse width on the surface texture of polyoxymethylene. Higher laser power combined with lower scanning speed facilitated the formation of deeper textures. Surface roughness increased with scanning speed, and with increasing laser power it first increased and then decreased. Su [33] optimized a hybrid laser cleaning process using the response surface methodology (RSM). The results showed that scanning speed had a significant effect on surface roughness, with higher scanning speeds leading to reduced roughness.

Computational fluid dynamics (CFD) simulation can provide high-fidelity velocity and temperature fields of the molten pool during welding [34], making it an essential tool for studying flow behavior in the welding process. Andreas [35] used a high-precision numerical model to investigate the significant influence of melt vaporization in non-penetration welding. Li [36] studied the effect of adjustable-ring-mode lasers on keyhole behavior and pore formation during non-penetration welding. Numerical simulation results showed that adjustable-ring-mode lasers can effectively enlarge the keyhole opening, improve keyhole stability and reduce pore formation. Zhang [37] developed a three-dimensional numerical model that integrates the volume of fluid (VOF) method with a ray-tracing algorithm and extended it from laser non-penetration welding of thick plates to penetration welding of 10 mm thick low-carbon steel. Hu [38] established a numerical model for laser penetration welding of 20 mm thick stainless steel. However, most existing numerical models for thick-plate butt welding are constructed in the form of flat-plate surfacing, ignoring the effects of butt joint gaps and the surface condition of the joint cross section. These factors are typically only incorporated when the influence of preset gaps is the primary research focus. For example, Guo [39] investigated the effect of various gap sizes on weld porosity by modeling butt joint gaps of 8 μm , 12 μm , and 16 μm .

Ai [40] introduced a 0.1 mm gap in the numerical model for laser welding of T-joints.

However, when cross-sectional microstructuring is introduced, the surface condition and gap at the joint cross section have a significant impact on the welding process and cannot be simplified or neglected in modeling.

To reduce process complexity, this study proposes a method of cross-sectional surface microstructuring to suppress hump defects formation in ultra-high-power laser penetration welding of thick plates. First, a pulsed laser is used to create surface microstructure on the butt joint cross section. Welding experiments are then conducted under identical laser welding parameters. Subsequently, a numerical model incorporating the microstructured surface of the joint cross section is developed. By combining high-speed imaging of molten pool flow with numerical simulation results, the mechanism by which the surface microstructure suppresses hump defects is revealed. Finally, differences in weld microstructure before and after surface microstructuring are compared, and tensile properties are evaluated.

2. Experiment and numerical model

2.1. Materials

The experimental material used was 20 mm thick 316L austenitic stainless steel, with elemental composition shown in Table 1. The welding experiments were conducted using a butt joint configuration, joining two specimens each measuring 200 mm in length and 50 mm in width. Prior to welding, the butt joint surfaces were milled flat and cleaned with alcohol.

Table 1. Chemical compositions of 316L.

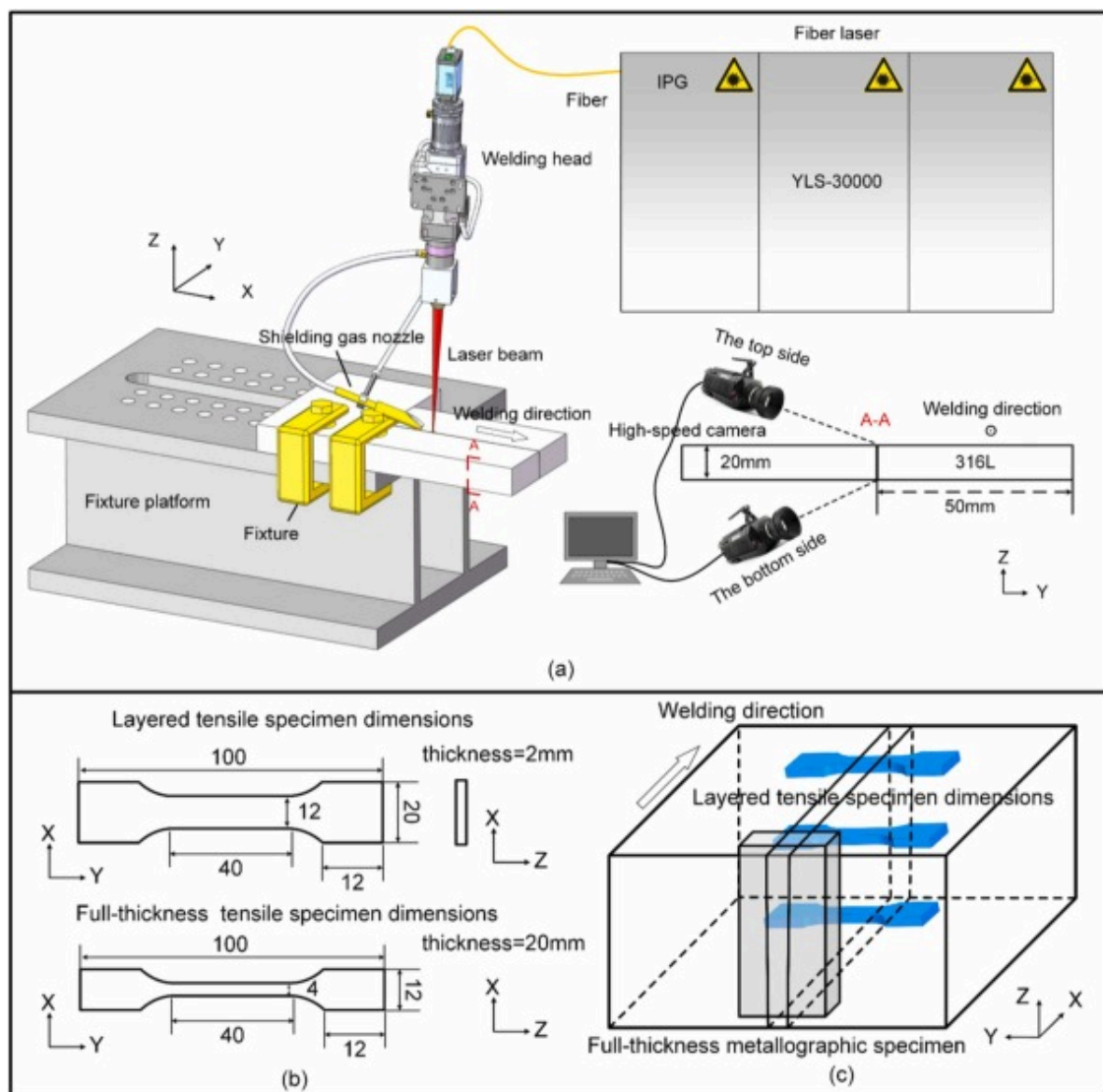
Element	C	Cr	Ni	Si	Mn	Mo	P	S	Fe
316L	0.024	16.69	11.64	0.54	1.25	2.45	0.024	0.016	Bal.

2.2. Welding set-up

Based on extensive preliminary process trials [38], welding parameters prone to hump defects formation were selected: laser power of 20,000 W, welding speed of 1.5 m/min, and zero defocus. Five sets of experiments listed in Table 2 were conducted using these parameters, with P-0 serving as the control group without any microstructuring of the butt joint cross section. First, surface microstructure was fabricated on the butt joint surfaces using a pulsed laser according to the process parameters of P-1 to P-4 in Table 2. Subsequently, the two specimens were spot welded together and clamped on the fixture platform as shown in Fig. 1. Finally, welding experiments were performed under the same welding parameters.

Table 2. Process parameters for surface microstructuring of butt joint cross-section.

	Laser power	Scanning speed	Pulse width	Pulse frequency	Scanning gap
Number	(W)	(mm/s)	(ms)	(Hz)	(mm)
P-0	0	0	0	0	0
P-1	45	150	0.1	1000	0.2
P-2	90	100	0.1	1000	0.2
P-3	120	75	0.1	1000	0.4
P-4	120	50	0.1	1000	0.8



Download: [Download high-res image \(353KB\)](#)

Download: [Download full-size image](#)

Fig. 1. Schematic diagrams of the welding experiment setup and sample preparation: (a) Welding experiment platform; (b) Schematic of tensile specimen dimensions; (c) Sampling locations for layered tensile specimens and metallographic samples.

Based on existing literature [28], [33] and preliminary experimental experience, a grid-shaped scanning strategy was selected, and laser power and scanning speed were identified as the key factors affecting texture depth and surface roughness. Excessively high laser power may cause over-melting, leading to molten pool instability and the accumulation of resolidified material around pits, which ultimately results in Ra stabilizing or even decreasing due to over-melting and smoothing effects. Excessively low scanning speed greatly reduces processing efficiency. Conversely, very high scanning speed or too low laser power can reduce the linear energy density below the evaporation threshold, resulting in shallower and smaller pits and lower surface Ra. The scanning gap controls the horizontal distance between adjacent scanning paths and determines the overlap between pits formed in the vertical scanning direction. The interaction between adjacent scanning paths depends on the ratio of the scanning gap to the laser spot diameter. To ensure the formation of discrete pits rather than continuous grooves along a single scan line, the scanning gap must be significantly larger than the laser spot diameter. Preliminary experiments showed that when the scanning gap exceeded 0.8 mm, the proportion of untextured surface began to dominate, causing the Ra to remain constant or slightly decrease. To achieve the core objective of suppressing welding hump defects, the influence of different roughness levels on the butt joint surface was systematically investigated. Based on this, the surface processing parameters listed in Table 2 were carefully designed to precisely produce samples with varied surface morphologies, providing a reliable experimental foundation for subsequent studies on the suppression of welding hump defects.

Surface microstructuring was performed using a Raycus RFL-QCW150/1500 fiber laser emitting radiation at a wavelength of 1080 nm. This laser system can switch between pulsed and continuous modes, with a maximum average power of 150 W in pulsed mode. The laser output head is a QBH-type connector. The focused diameter of the output laser is 160 μm . The welding experiments were conducted using an IPG-30000 fiber laser with an output wavelength of 1070 nm. The collimating length and focusing length of the Scansonic BO/SF welding head are 175 mm and 350 mm, respectively. The fiber core

diameter is 300 μm , and the focused spot diameter is 600 μm . The laser head is mounted on a KUKA KR120 robotic arm. As shown in Fig. 1, the welding torch was tilted 5° relative to the welding direction, and a trailing shielding gas nozzle was positioned behind the welding direction to protect the upper surface of the molten pool with argon gas of 99.9 % purity. High-speed cameras captured the molten pool flow behavior from both the upper and backside with image intervals of 0.2 ms.

2.3. Microstructure characterization and mechanical testing analysis

After welding, the weld morphology on both the upper and backside of the specimens was first captured using a high-resolution scanner. Then, full-thickness metallographic samples of the weld and unwelded sections at the tack-welded positions were prepared using wire electrical discharge machining. As shown in Fig. 1, full-thickness tensile specimens were also fabricated to evaluate the overall tensile performance of the joints. Full-thickness tensile specimens refer to samples that retain the weld formation on both the upper and bottom surfaces of the weld, enabling them to reflect the actual service performance of the joint. To investigate the microstructural and mechanical property variations at different weld depths and their correlation, 2 mm thick layered tensile specimens were prepared at 2 mm, 10 mm, and 18 mm below the weld upper surface. Corresponding weld microstructures were observed at the same locations on the full-thickness metallographic samples and designated as UZ (upper zone), MZ (middle zone), and LZ (lower zone), respectively.

The morphology and surface roughness of the microstructured butt joint cross section were measured using a laser confocal scanning microscope (CLSM, VK-X200K, KEYENCE CO., LTD., Japan). The wire-cut metallographic samples were then ground and polished, followed by electrolytic polishing of their surfaces in a 10 % perchloric acid ethanol solution. The polishing parameters were set at 20 V for 30 s. Electron backscatter diffraction (EBSD) data of the weld microstructure were acquired using a GeminiSEM300 field emission scanning electron microscope (Carl Zeiss, Germany) with a scanning step size of 2 μm . Tensile tests were conducted on a Shimadzu AG-100kN testing machine at a

strain rate of 5 mm/min, with three repetitions performed at each position. Finally, fracture surface morphologies were observed using the SEM.

3. Mathematical models

3.1. Assumptions

Laser welding of metals involves the absorption of laser energy by the material, followed by phase transitions such as melting, vaporization, and solidification, as well as ionization of metal vapor. It is a complex physicochemical process. When developing a numerical model, it is essential to strike a balance between model accuracy, computational cost and efficiency. This challenge becomes more pronounced in thick-plate welding simulations, where the plate thickness (≥ 20 mm) is several times greater than that of conventional models, requiring significantly more computational grids and solution time. Therefore, the following reasonable assumptions are made to simplify the numerical model [41], [42].

1. The molten pool flow is assumed to be laminar, and the liquid metal is treated as an incompressible Newtonian fluid.
2. The metal is assumed to be isotropic and homogeneous.
3. The initial temperature of the model is set to 293 K, and the influence of plasma on the welding process is neglected.

3.2. Heat source in CFD model

In a stable optical cavity, the transverse profile of the optical field remains constant at any position along the propagation direction of the optical axis, although the beam diameter can vary. The laser heat source is modeled as a Gaussian beam, and the variation of the effective beam radius $w(z)$ at a distance z from the beam waist along the propagation

direction follows a hyperbolic trend [43]:

$$\left(\frac{w(z)}{w_0}\right)^2 - \left(\frac{\lambda z}{\pi w_0^2}\right)^2 = 1 \quad (1)$$

Here, λ is the laser wavelength, and w_0 is the beam radius at the waist. According to Eq. (1), the Rayleigh length can be calculated as $z_R = \frac{\pi w_0^2}{\lambda}$, which means that at a distance of $\frac{\pi w_0^2}{\lambda}$ along the propagation direction from the beam waist, the beam cross-sectional area doubles compared to that at the waist. The intensity distribution of a Gaussian beam at a cross-section can be expressed as [41]:

$$I(x, y, z) = \frac{3Q}{\pi w^2(z)} e^{-\frac{3(x^2+y^2)}{w^2(z)}} \quad (2)$$

x, y and z represent the three orthogonal axes in the Cartesian coordinate system, and Q denotes the laser power.

3.3. Governing equations

The foundation of FLOW-3D simulation lies in solving the equations that describe the fundamental physical laws of fluid motion, namely the Navier–Stokes equations and the continuity equation for incompressible or weakly compressible flows. These equations are discretized in three dimensions, and the velocity and pressure fields are solved within each cell of the computational mesh [44], [45].

Mass conservation equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \vec{v} = 0 \quad (3)$$

Where, ρ is fluid density, t is time, $\vec{v}$ is velocity vector.

Momentum conservation equations:

$$\rho \frac{\partial \vec{v}}{\partial t} + \rho \nabla \cdot (\vec{v} \otimes \vec{v}) = -\nabla P + \mu \nabla^2 \vec{v} + \vec{f}_b - \frac{\rho \mu}{K} \vec{v} \quad (4)$$

Where, P is ambient pressure, μ is dynamic viscosity, K is porous media permeability.

Energy conservation equation:

$$\rho \frac{\partial H}{\partial t} + \rho \nabla \cdot (\vec{v} H) = \nabla \cdot (k \nabla T) + q \quad (5)$$

Within the energy conservation framework, H denotes enthalpy, k represents material thermal conductivity, T indicates temperature, and q designates external energy source terms.

3.4. Solidification and phase change model

During molten pool solidification, the progressive increase in solid fraction within the mushy zone induces momentum dissipation in flowing liquid metal. To address this, the enthalpy-porosity technique models the solid-liquid interface by treating the mushy region as an isotropic porous medium [46], where the local porosity equates to the liquid volume fraction.

Porous media permeability (K) is derivable from the Karman-Kozeny equation:

$$K = \frac{f_l^3 + \delta_0}{A_m (1 - f_l)^2} \quad (6)$$

Here, δ_0 denotes a non-zero constant scaling factor, while A_m represents the Darcy coefficient for the mushy region. The liquid phase volume fraction f_l follows the established formulation [47]:

$$f_l = \begin{cases} 0 & T < T_s \\ \frac{T - T_s}{T_l - T_s} & T_s < T < T_l \\ 1 & T > T_l \end{cases} \quad (7)$$

Here, T_s and T_l denote the solidus and liquidus temperatures, respectively. In the solid phase region, $f_l = 0$; in the liquid phase region, $f_l = 1$; and in the mushy region, $0 < f_l < 1$.

The volume-of-fluid (VOF) method is employed, in which a scalar function called the volume fraction is introduced. Within each computational cell, a convection transport equation is solved to determine the volume fraction at the next time step, thereby enabling the tracking of the free surface of the molten pool [48].

$$\frac{\partial f_{VOF}}{\partial t} + \nabla \cdot (v f_{VOF}) = 0 \quad (8)$$

The phase volume fraction f_{VOF} quantifies liquid metal occupancy within a control volume. $f_{VOF} = 0$ indicates a purely gaseous region, $f_{VOF} = 1$ corresponds to molten metal saturation, and intermediate values ($0 < f_{VOF} < 1$) locate the liquid-gas interface.

3.5. Multi-reflection mode

During thick-plate laser welding, keyhole formation is inevitable. Within this cavity, multireflection of the laser beam intensifies energy density at the keyhole base, enhancing its depth and acuity. Our CFD model incorporates a Fresnel-absorption-based multireflection algorithm, enabling high-fidelity simulation of keyhole evolution and molten pool dynamics in deep-penetration welding.

The reflected ray $\mathbf{x}_r$ can be expressed as [49], [50]:

$$\mathbf{x}_r = \mathbf{x}_i - 2(\mathbf{x}_i \cdot \mathbf{n}) \quad (9)$$

Defining $\mathbf{x}_i$ as the incident ray direction and $\mathbf{n}$ as the surface normal vector, the laser undergoes multiple Fresnel reflections within the keyhole until energy depletion via absorption or emission from the cavity.

In Fresnel absorption based laser deep penetration welding, the absorption rate is expressed by the following equation [42], [51]:

$$a = 1 - \frac{1}{2} \left(\frac{1 + (1 - \varepsilon \cos \alpha)^2}{1 + (1 + \varepsilon \cos \alpha)^2} + \frac{\varepsilon^2 - 2\varepsilon \cos \alpha + 2\cos^2 \alpha}{\varepsilon^2 + 2\varepsilon \cos \alpha + 2\cos^2 \alpha} \right) \quad (10)$$

Where α denotes the incidence angle between the laser ray and surface normal vector $\mathbf{n}$,

while ε represents a material-laser dependent coefficient [52]:

$$\varepsilon^2 = \frac{2\varepsilon_2}{\varepsilon_1 + \left[\varepsilon_1^2 + \left(\frac{\sigma_{st}}{\omega \varepsilon_0} \right)^2 \right]^{\frac{1}{2}}} \quad (11)$$

Where, ε_{st} is the electric conductivity depth per unit (base material), ε_2 is relative permittivity (base material), ε_1 is relative permittivity(plasma), ε_0 is permittivity of vacuum, ω is laser regular frequency. The value of ε for this model is taken as 0.10.

3.6. Driving force

During the welding process, several key driving forces exist, including recoil pressure P_R , surface tension P_s , buoyancy f_b , protective gas dynamic pressure P_D , and gravity. These forces play crucial roles in the molten pool flow and the evolution of the keyhole shape, ultimately determining the weld formation quality.

Recoil pressure is the reactive force exerted on the molten metal during vaporization caused by laser irradiation. It is the primary driving force for keyhole penetration and expansion. The magnitude of recoil pressure is positively correlated with temperature and can be expressed as follows [53]:

$$P_R(T) = \beta P_{sat} = \beta P_v \exp \left\{ \frac{\Delta H_v}{(\tau - 1)c_v T_v} \left(1 - \frac{T_v}{T} \right) \right\} \quad (12)$$

where, β is evaporation coefficient, $0 \leq \beta \leq 1$; P_{sat} is evaporation pressure, P_v is saturation pressure, ΔH_v is evaporation latent heat, τ is the ratio of the vapor specific heats, c_v is steam constant specific heat and T_v is saturation temperature.

Surface tension force (P_s) within the molten pool drives melt contraction toward energy minimization, exhibiting temperature-dependent attenuation that predominantly facilitates keyhole shrinkage or collapse. This thermo-capillary effect follows:

$$P_s = \gamma + A_\gamma (T - T_l) \quad (13)$$

where, γ represents the coefficient of the surface tension A_γ represents the temperature gradient of the surface tension.

During laser deep-penetration welding, the gas shielding effect above the molten pool is implemented in this CFD model by applying dynamic pressure P_D at the melt surface.

$$P_D = \frac{1}{2} \rho_{gas} v_{gas}^2 \cos \alpha_0 \quad (14)$$

where ρ_{gas} denotes shielding gas density, v_{gas} is the gas velocity, and α_0 the incidence angle between the keyhole surface normal and gas flow direction vectors.

The Boussinesq approximation models thermal buoyancy f_b induced by temperature gradients. Thermal buoyancy f_b in the same direction as the acceleration of gravity. Therefore, it is only added to the z-direction momentum conservation equation:

$$f_b = \rho_0 \beta g (T - T_0) \quad (15)$$

Where, T_0 is the ambient temperature, ρ_0 is the density of the fluid at T_0 , β is the coefficient of thermal expansion, and g is the acceleration of gravity.

3.7. Boundary conditions and simulation cases

Thermal boundary conditions for thick plate [54]:

$$k \frac{\partial T}{\partial n} = -h_c (T - T_{ref}) - \varepsilon_r \sigma (T^4 - T_{ref}^4) \quad (16)$$

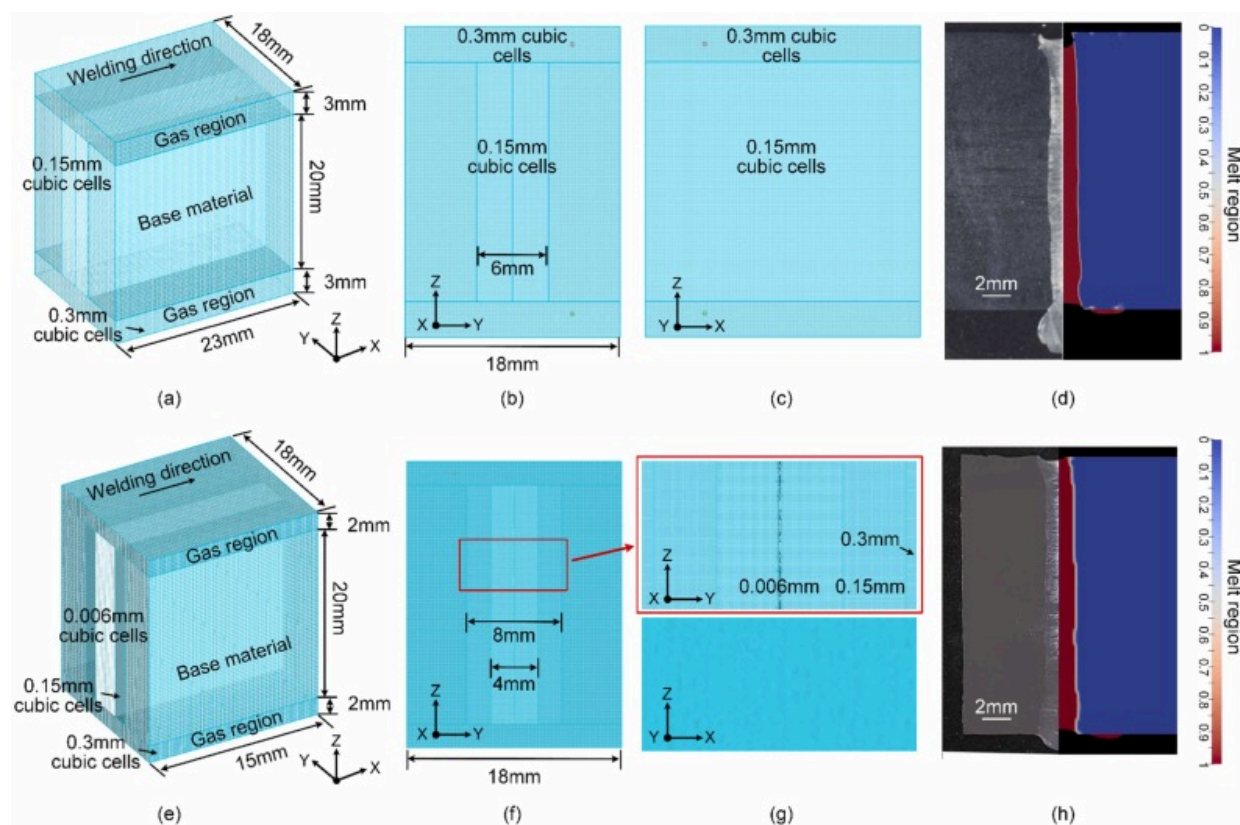
where, n is the normal vector at the free interface, h_c the convective heat transfer coefficient, ε_r the material surface emissivity, σ is the Stephan-Boltzmann constant, T_{ref} is the ambient temperature.

Free surface of the keyhole [48]:

$$k \frac{\partial T}{\partial n} = q_{abs} - \varepsilon_r \sigma (T^4 - T_{ref}^4) - h_c (T - T_{ref}) - W_{evp} H_v \quad (17)$$

where, q_{abs} is the absorbed laser energy, W_{evp} and H_v are the evaporation rate and the latent heat of evaporation of the liquid metal.

As shown in Fig. 2a, a conventional numerical model for a 20 mm thick plate was established using FLOW-3D fluid simulation software. The simulation time was set to 0.8 s, covering a welding length of 20 mm. The central 20 mm thickness in the model corresponds to the welded specimen, while the 3 mm thicknesses at the top and bottom represent the gas domain. The mesh size was set to 0.3 mm. To balance computational efficiency and accuracy, mesh refinement was applied at the weld zone with a fine mesh size of 0.15 mm. The mesh resolution was previously validated in earlier work [38]. The total number of mesh cells in the model was 1.37 million, and the solving time was 13 h.



Download: [Download high-res image \(687KB\)](#)

[Download: Download full-size image](#)

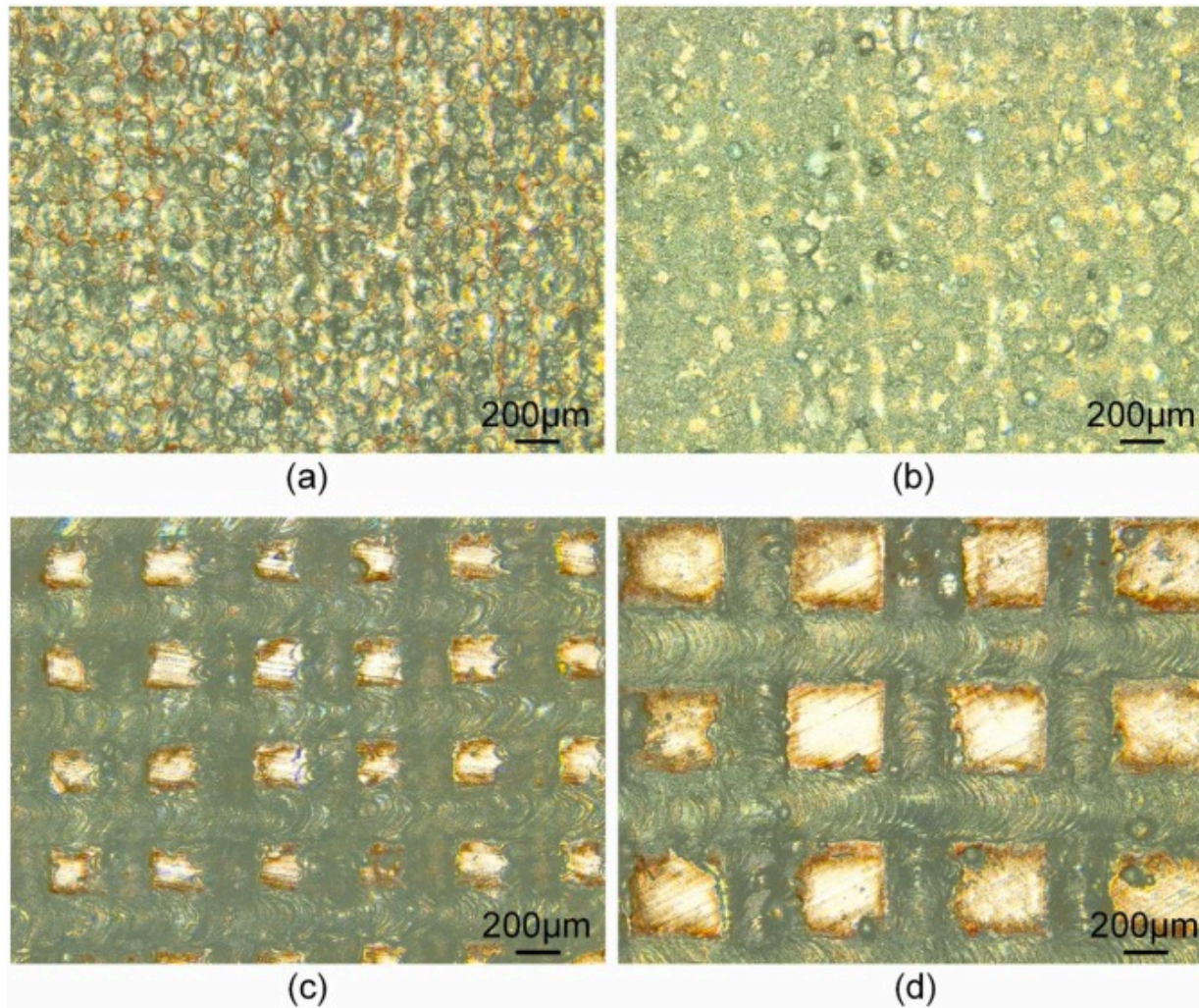
Fig. 2. Schematic diagrams of numerical models and model validation results: (a) Conventional numerical model for welding 20 mm thick plate; (b) Mesh division in Y-Z view; (c) Mesh division in X-Z view; (d) Comparison between experimental metallographic cross section and simulation results; (e) Numerical model for welding 20 mm thick plate incorporating butt joint surface morphology; (f) Mesh division in Y-Z view; (g) Mesh division in X-Z view; (h) Comparison between experimental metallographic cross section and simulation results.

To investigate the influence of the butt joint surface microstructure on weld formation, a 20 mm thick plate welding numerical model incorporating the joint surface microstructure was established, as shown in Fig. 2e. The laser confocal microscopy data were first converted into an STL file. Since the original confocal scan covered an area of 2 mm × 2.9 mm, the surface microstructure data were tiled to the required modeling size of 15 mm × 20 mm and then imported into FLOW-3D software. The upper image in Fig. 2g shows the butt joint morphology in the model with a certain gap, while the lower image presents the joint cross-section microstructure from the X–Z view, where wrinkles caused by the surface microstructure are visible. Mesh refinement to 0.006 mm was applied in the weld zone to preserve the surface microstructure of the butt joint without distortion. The welding simulation time was set to 0.5 s. The total mesh count was 6.16 million, and the solving time was 6 days and 10 h. The material physical properties used in the model are listed in Table 3. The material property parameters used in the numerical model were obtained from the FLOW-3D material database “MaterialsDatabase.”(Fig. 3)

Table 3. Thermal physical properties of 316 L stainless steel.

Name	Symbol	Unit	Value
Solidus density	ρ_s	kg m ⁻³	7950
Liquidus density	ρ_l	kg m ⁻³	6881

Name	Symbol	Unit	Value
Thermal conductivity of solid	k_s	$\text{W m}^{-1} \text{K}^{-1}$	13
Thermal conductivity of liquid	k_l	$\text{W m}^{-1} \text{K}^{-1}$	29
Specific heat capacity of solid	C_{ps}	$\text{J kg}^{-1} \text{K}^{-1}$	722
Specific heat capacity of liquid	C_{pl}	$\text{J kg}^{-1} \text{K}^{-1}$	830
Latent heat of fusion	H_m	J kg^{-1}	2.6×10^5
Latent heat of evaporation	H_v	J kg^{-1}	6.08×10^6
Coefficient of surface tension	γ	N m^{-1}	1.804
Temperature gradient of surface tension	A_γ	$\text{N m}^{-1} \text{K}^{-1}$	-0.39×10^{-3}
Dynamic viscosity	μ	Pa s	0.08
Solidus temperature	T_s	K	1674
Liquidus temperature	T_l	K	1697
Convective heat transfer coefficient	h_c	$\text{W K}^{-1} \text{m}^{-2}$	10
Thermal emissivity	ε_r		0.8
Ambient temperature	T_{ref}	K	293



[Download: Download high-res image \(1MB\)](#)

[Download: Download full-size image](#)

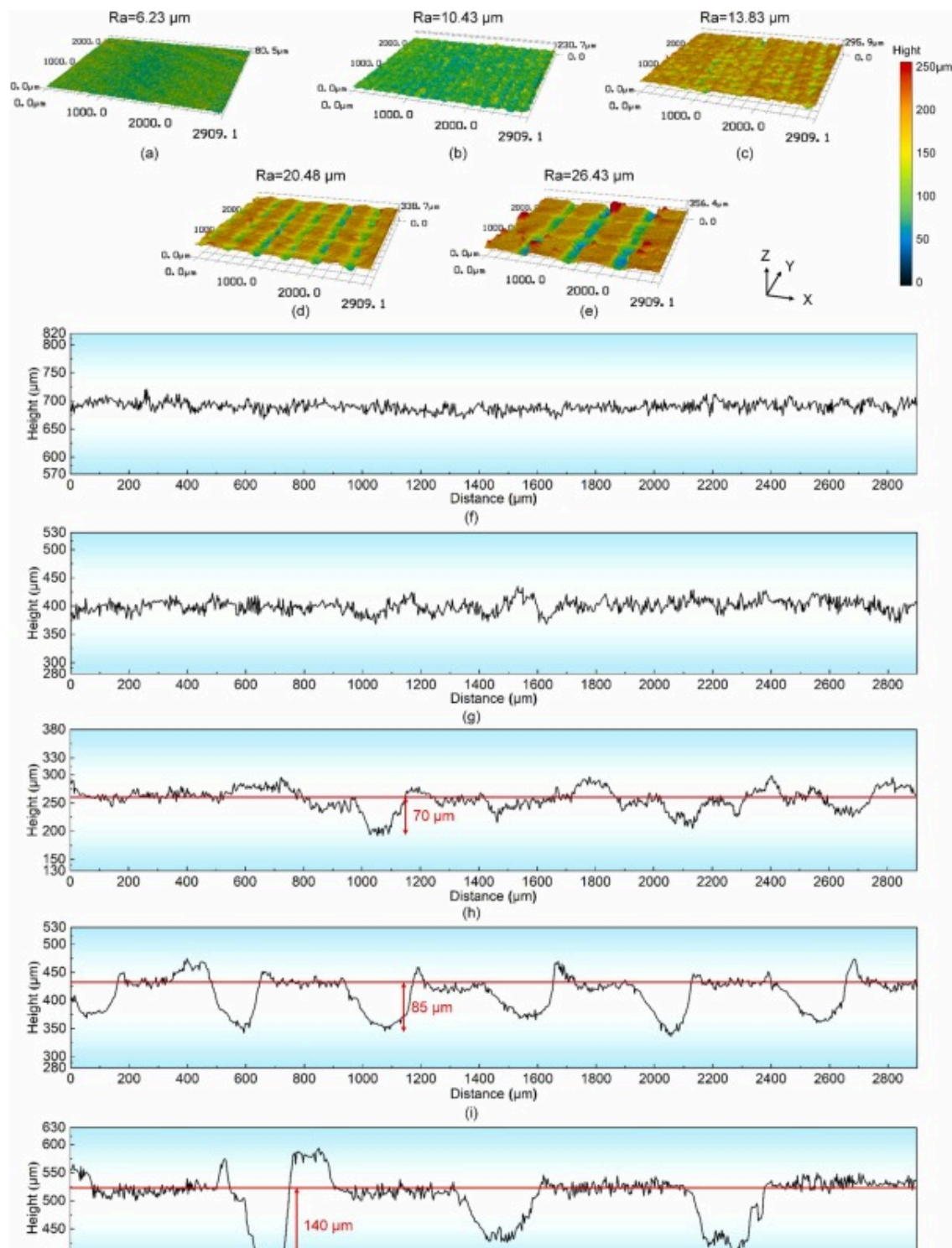
Fig. 3. Surface morphology of butt joint cross-sections after pulsed laser treatment: (a) Surface morphology after treatment with parameter P-1; (b) Surface morphology after treatment with parameter P-2; (c) Surface morphology after treatment with parameter P-3; (d) Surface morphology after treatment with parameter P-4.

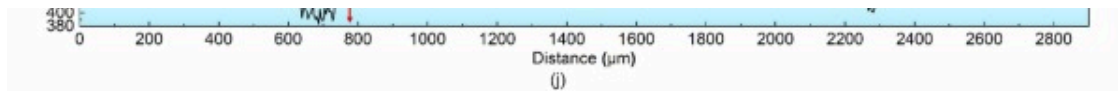
4. Results

4.1. Characterization of surface microstructure

To ensure consistent surface conditions at the butt joint cross-sections, all sections were subjected to milling pretreatment, followed by microstructuring of the milled surfaces using the process parameters listed in [Table 2](#). Under the P-1 parameter set, the use of low laser power and high scanning speed leads to insufficient connectivity between adjacent molten pool depressions. As a result, the processed surface presents an intersecting, fish-scale-like overlapping structure. Compared with the P-1 parameters, P-2 increases the laser power and decreases the scanning speed, enabling the molten pool pits to connect and form continuous ripple-like grooves, with an increased number of spherical micro-protrusions on the surface. Parameters P-3 and P-4 have larger scanning gaps, resulting in unmelted milled surfaces that clearly display a rectangular array distribution with sharp and distinct edges. Notably, the grooves processed under P-3 are free of residual spatter, whereas those treated with P-4 show scattered metallic spatter particles on both the grooves and the rectangular surface.

[Fig. 4a-e](#) show the confocal microscopy results of surfaces after milling and microstructuring, while [Fig. 4f-j](#) present the corresponding line profile curves along the X direction for the five surface conditions. The milled surface is relatively flat, with height variations within 50 μm . After treatment with parameter P-1, the surface exhibits an interlaced molten pool pit morphology, with height variations between 50 μm and 100 μm . Treatment with parameter P-2 results in deeper molten pool pits, with depths not exceeding 75 μm . Surfaces treated with parameters P-3 and P-4 show continuous grooves of greater depth, and metal particles formed by solidified spattered droplets appear elevated above the surface.





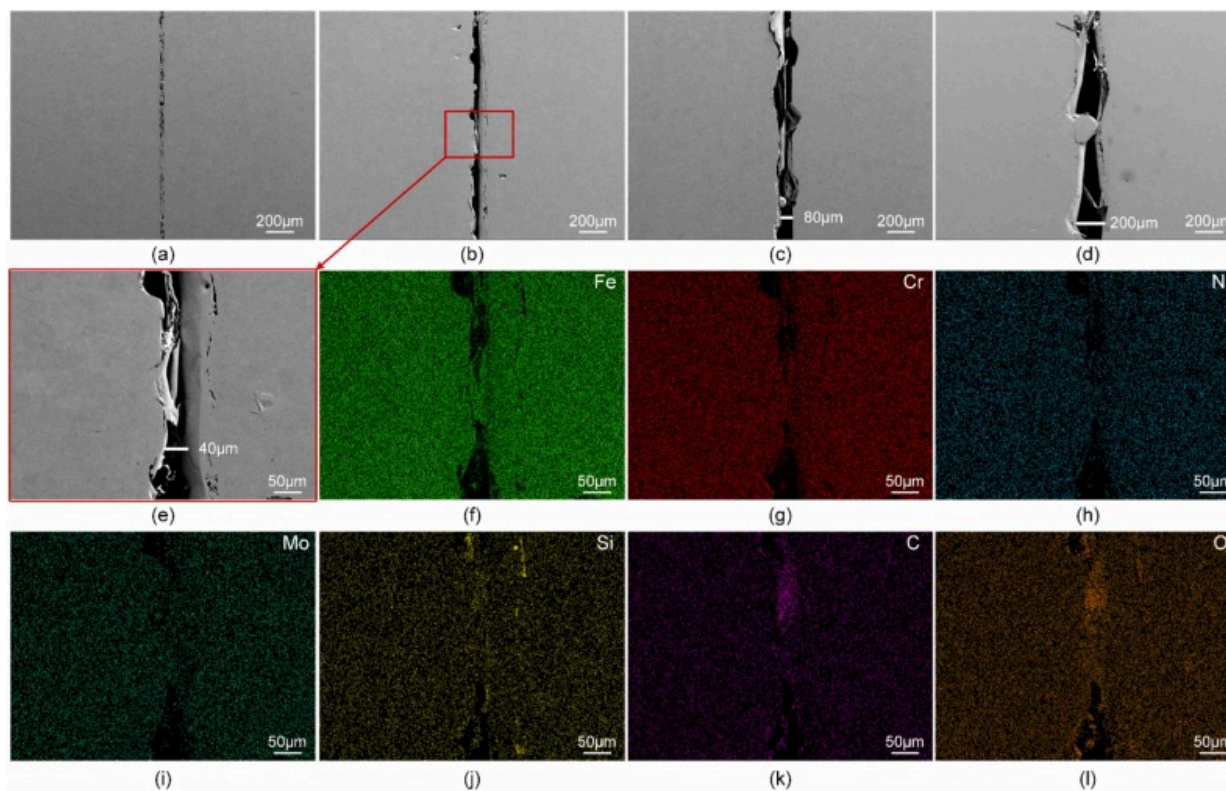
Download: [Download high-res image \(655KB\)](#)

Download: [Download full-size image](#)

Fig. 4. Confocal microscopy results and line roughness profiles of butt joint cross-sections after pulsed laser treatment: (a) Confocal image of the surface after milling; (b-e) Confocal surface results after treatment with parameters P-1, P-2, P-3, and P-4, respectively; (f) Line profile in the X direction after milling; (g-j) Line profiles in the X direction after treatment with parameters P-1, P-2, P-3, and P-4, respectively.

After welding, cross-sectional metallographic samples were cut at the A-A section shown in Fig. 6 to observe the butt joint condition of the plates after milling and surface microstructuring. Before welding, the two plates were fixed by spot welding, and the metallographic samples included the spot-welded area. Therefore, the metal blocks on both sides of the sample shown in Fig. 5a remain connected, and the gap between them reflects the initial butt joint condition of the two plates before welding. After treatment with parameter P-1, the surface microstructure of the two plates fits closely together, with the butt joint surfaces nearly in full contact. Following treatment with parameter P-2, the butt joint gap measures approximately 40 μm . No significant changes are observed in the content and distribution of the main alloying elements Cr, Ni, and Mo at the joint. The distributions of non-metallic elements such as Si, C, and O indicate that the flocculent structures in Fig. 5e and the black bands on the right surface are mainly residual embedding material within the gap, which do not affect the welding process. The butt joint gaps for parameters P-3 and P-4 increases to approximately 80 μm and 200 μm , respectively, and the surface undulations on one side are consistent with the line profiles shown in Fig. 4. In Fig. 5d, a protruding spherical metal particle on the left cross-section comes into contact with the right cross-section. This indicates that after pulsed laser treatment, the surface microstructure which consists of molten pool pits, grooves, and

protruding spherical metal particles, collectively determines the gap size and the initial condition of the butt joint cross-section.



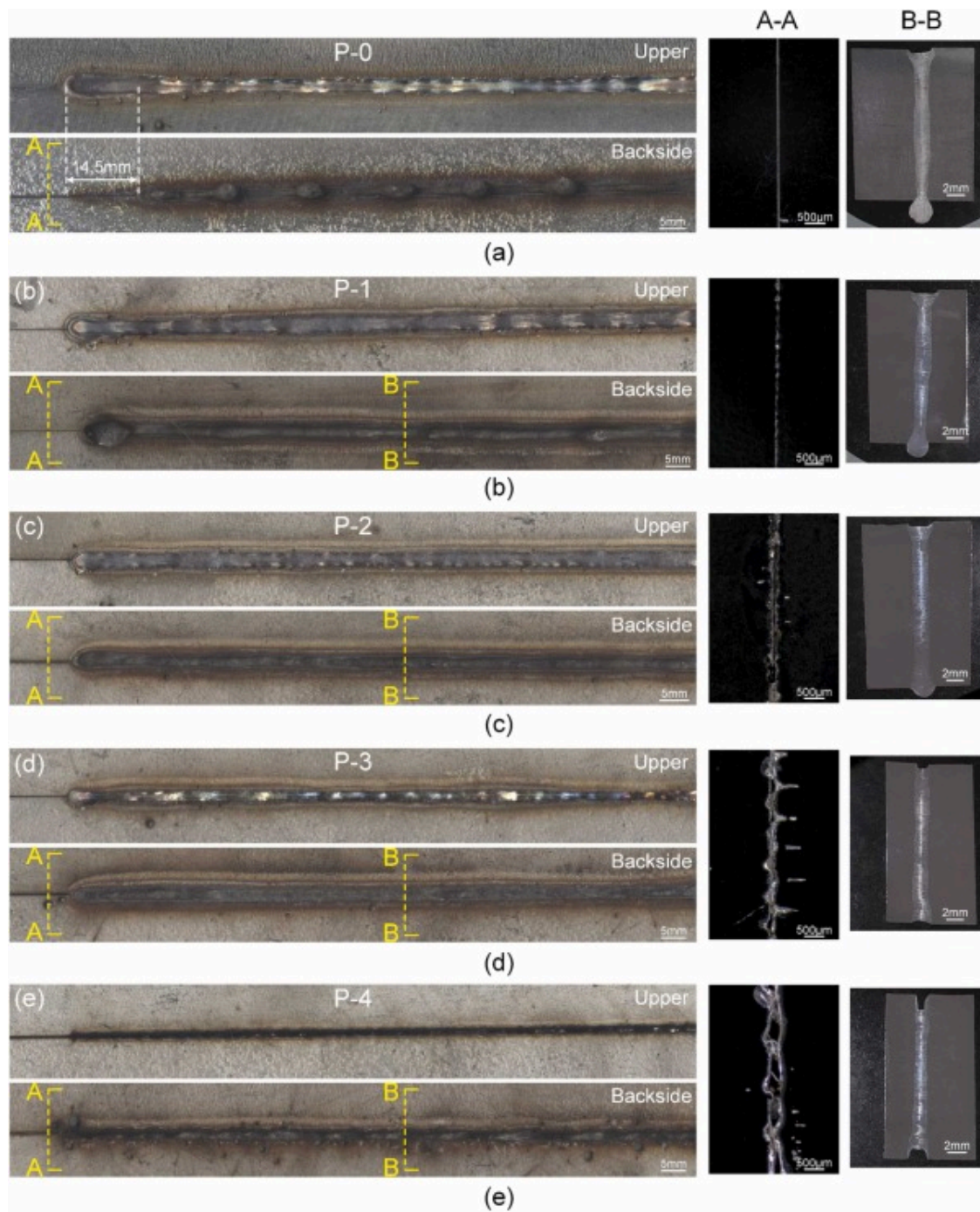
[Download: Download high-res image \(1MB\)](#)

[Download: Download full-size image](#)

Fig. 5. Initial cross-sectional state of the butt joint: (a) Cross-section after treatment with parameter P-1; (b) Cross-section after treatment with parameter P-2; (c) Cross-section after treatment with parameter P-3; (d) Cross-section after treatment with parameter P-4; (e) Locally magnified SEM image of (b); (f-l) Elemental distribution maps corresponding to (e).

4.2. Macroscopic formation

As shown in Fig. 6a, the butt joint cross-section of the control group P-0 was milled flat without surface microstructuring, resulting in nearly complete contact at the interface. After welding with parameter P-0, no obvious depression is observed on the upper surface of the weld, while uniformly sized and evenly distributed humps appeared on the backside. After treatment with parameter P-1, interlaced molten pool pits appeared on the butt joint cross-section, resulting in a slight increase in surface roughness compared with the milled surface. The undulating surface microstructure fits together closely, allowing the butt joint to maintain tight contact, as shown in Fig. 6b. No obvious depression is observed on the upper surface of the weld, while irregularly distributed protrusions are present on the backside. After treatment with parameter P-2, a certain butt joint gap is present, within which surface microstructure remains. The weld exhibits a more uniform formation on the upper surface, and the backside hump defects is significantly suppressed. In contrast, welding with parameter P-3 results in a pronounced depression on the upper surface and uneven weld formation. With parameter P-4, a distinct cutting-like morphology appears on the upper surface of the weld, while the backside exhibits a concave profile. These experimental results indicate that, under identical welding process parameters, introducing appropriate surface microstructure on the butt joint cross-section through pulsed laser treatment can effectively suppress the formation of hump defects.

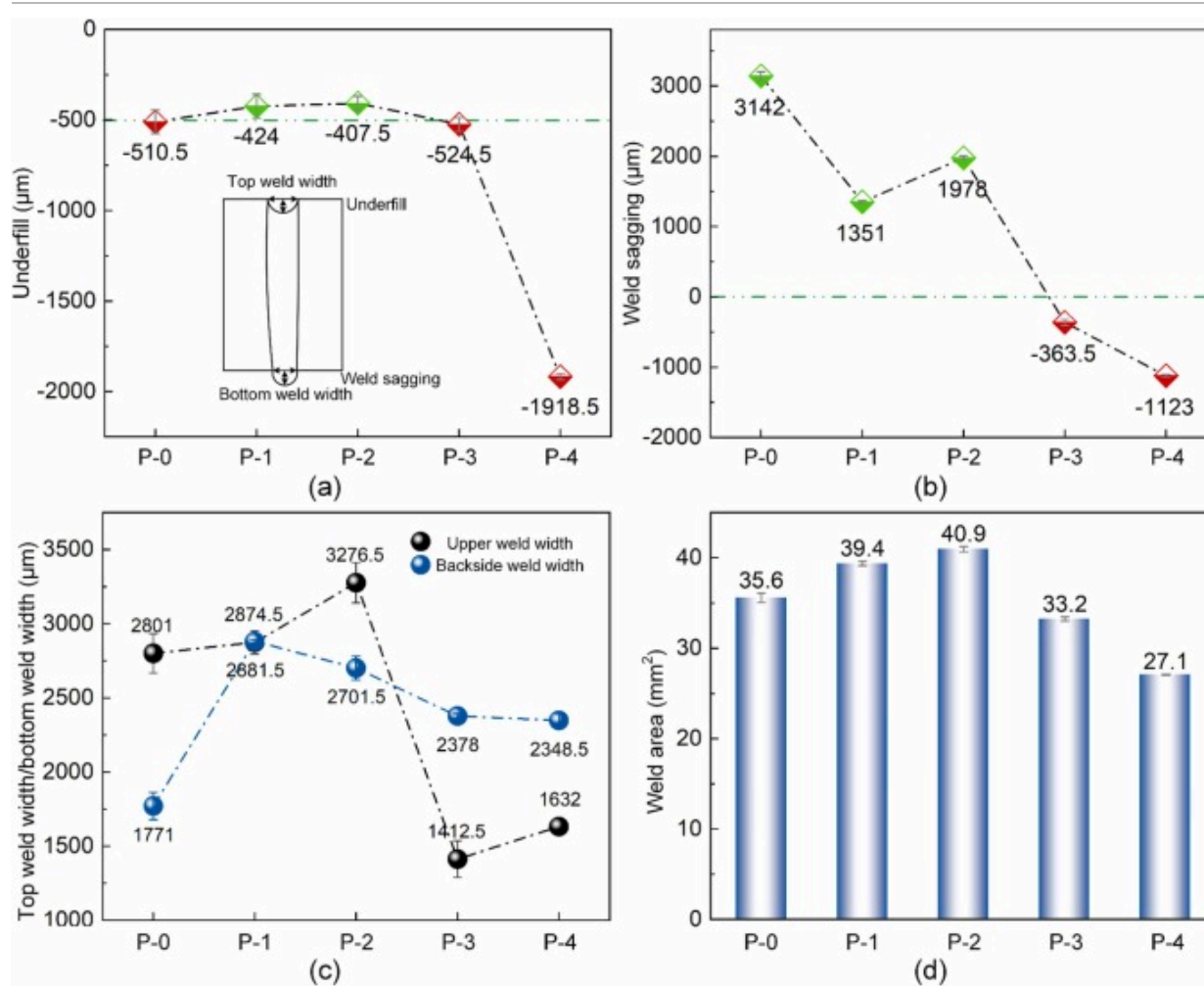


[Download: Download high-res image \(831KB\)](#)

[Download: Download full-size image](#)

Fig. 6. Weld surface morphology, initial butt joint cross-section condition, and cross-sectional metallographic results: (a) Control group P-0 without surface microstructuring, showing hump defects; (b) Weld formation after treatment with parameter P-1; (c) Weld formation after treatment with parameter P-2; (d) Weld formation after treatment with parameter P-3; (e) Weld formation after treatment with parameter P-4.

As illustrated in the schematic of [Fig. 7a](#), the characteristic geometric data of the weld seam are statistically analyzed. Each data point is obtained by measuring the same metallographic sample three times, with error bars representing the variation. According to GB/T 22085.1–2008 / ISO 13919–1:1996, the degree of underfill on the upper surface and the degree of sagging on the backside are measured. The strictest quality level, "strict B," sets the maximum allowable underfill at 500 μm . In [Fig. 7a,b](#), green dots indicate values that meet the standard requirements, while red dots indicate non-compliance. Based on the weld surface formation results, it can be concluded that treatment with parameter P-2 effectively suppressed the hump defects, resulting in good weld formation. Analysis of the weld width on the upper and backside surfaces shows that after surface microstructuring, the upper surface width first increases and then decreases as roughness increases. This indicates that the surface microstructure of the butt joint enhances energy absorption during the initial stage of laser interaction; however, excessive roughness (as seen with P-3 and P-4) adversely affects energy absorption. The trend in the molten cross-sectional area of the weld further supports this analysis. Furthermore, in the four microstructured groups, the backside weld width decreases with increasing roughness but consistently exceeds that of the untreated (P-0) sample.



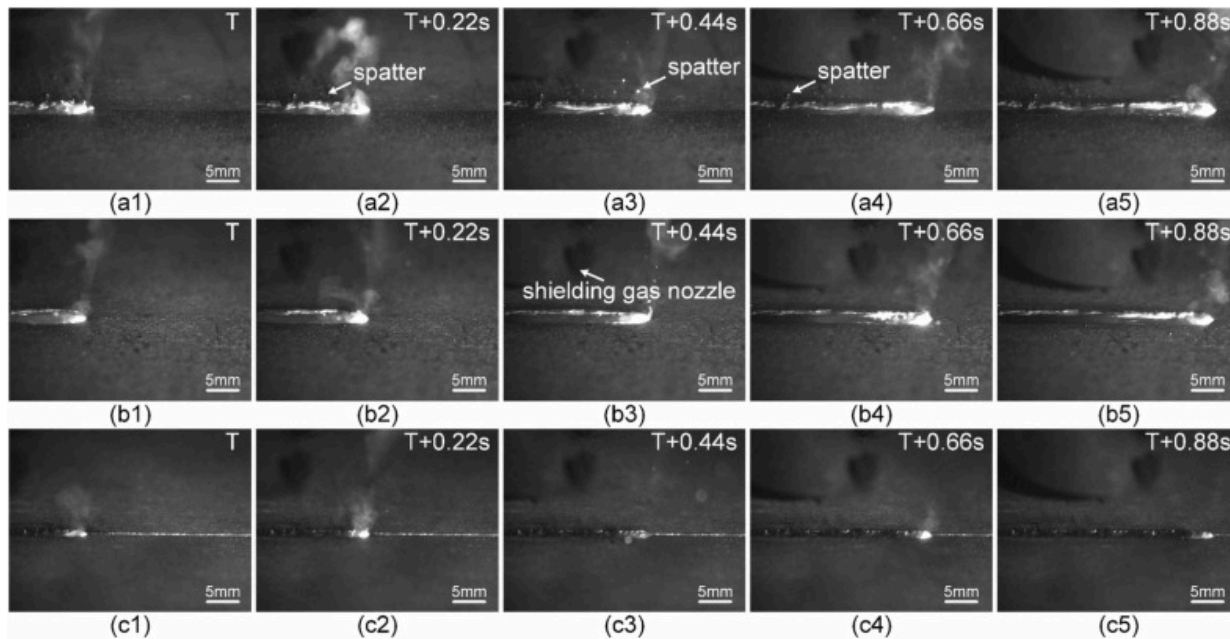
Download: [Download high-res image \(339KB\)](#)

Download: [Download full-size image](#)

Fig. 7. Characteristic dimensions of welds after surface microstructuring. Error bars indicated a 95 % confidence interval (CI): (a) Underfill on the weld upper surface; (b) Sagging on the weld backside; (c) Weld width at the upper and backside; (d) Variation of weld cross-sectional area.

4.3. Keyhole condition and molten pool flow behavior

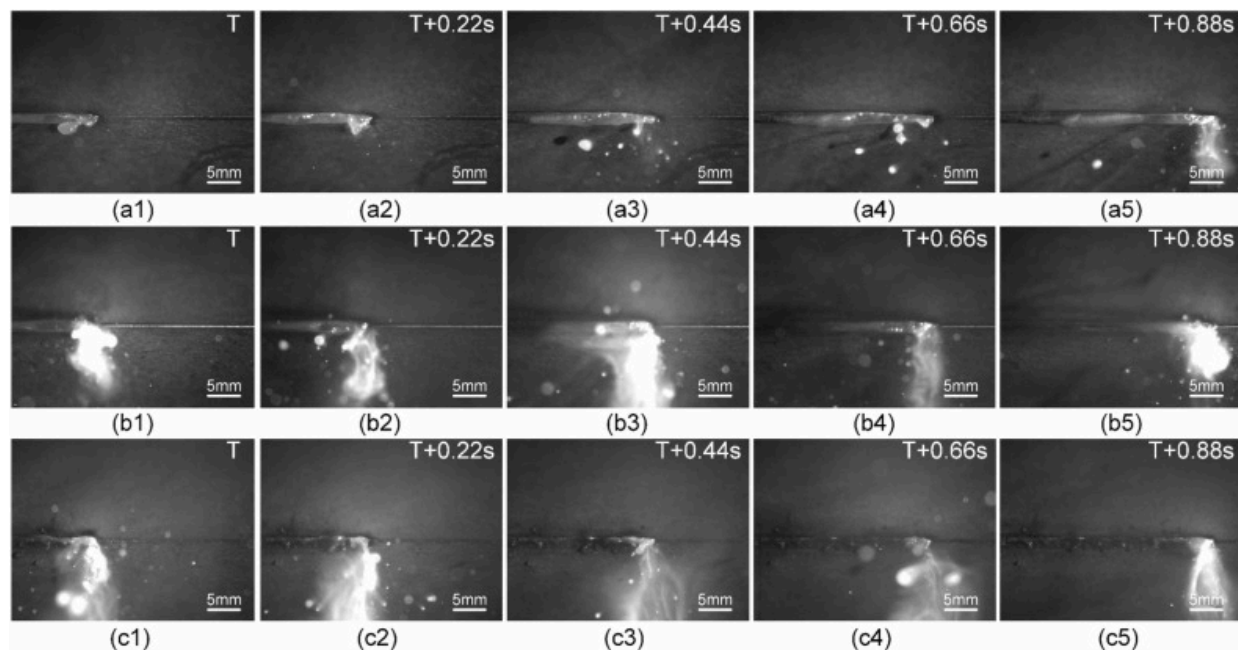
High-speed imaging was conducted during the welding process for three parameter sets: P-0, P-2, and P-4. As shown in Fig. 8a2, the molten pool on the upper surface of the weld under the P-0 condition appears slightly concave overall. During molten pool flow, slight instability is observed along with the occurrence of minor spatter. In the five time frames shown in Fig. 9a1-a5, the keyhole remains nearly closed throughout. The molten pool at the bottom exhibits unstable flow, accompanied by the detachment of small spherical molten droplets from the pool. For the hump defects suppressing parameter set P-2, the molten pool on the upper surface exhibits stable flow without any spatter. The overall flow of the molten pool is clearly visible, and no concavity is observed. In the five time frames shown in Fig. 9b1-b5, the keyhole remains in a penetrated state, accompanied by partial metal plume and a small amount of spatter. The flow at the bottom of the molten pool is uniform and stable, corresponding well with the weld backside formation shown in Fig. 6c. For parameter set P-4, the roughness of the butt joint cross section reaches $R_a = 26.43 \mu\text{m}$, and the joint gap is approximately $200 \mu\text{m}$, which is about one-third of the laser beam's focal spot diameter. The molten pool on the upper surface appears concave with cutting-like morphology and the weld width varies significantly along the welding direction. As shown in the high-speed imaging result in Fig. 9c4, the keyhole remains in a fully penetrated state. A large amount of molten material is ejected from the keyhole, with some detaching directly from the molten pool and some being expelled to both sides of the weld root.



[Download: Download high-res image \(356KB\)](#)

[Download: Download full-size image](#)

Fig. 8. High-speed imaging results of the weld upper surface: (a1-a5) Parameter set P-0 with hump defects; (b1-b5) Parameter set P-2 with suppressed hump defects; (c1-c5) Weld with butt joint treated using parameter P-4.



Download: [Download high-res image \(314KB\)](#)

Download: [Download full-size image](#)

Fig. 9. High-speed imaging results of the weld backside: (a1-a5) Parameter set P-0 with hump defects; (b1-b5) Parameter set P-2 with suppressed hump defects; (c1-c5) Weld with butt joint treated using parameter P-4.

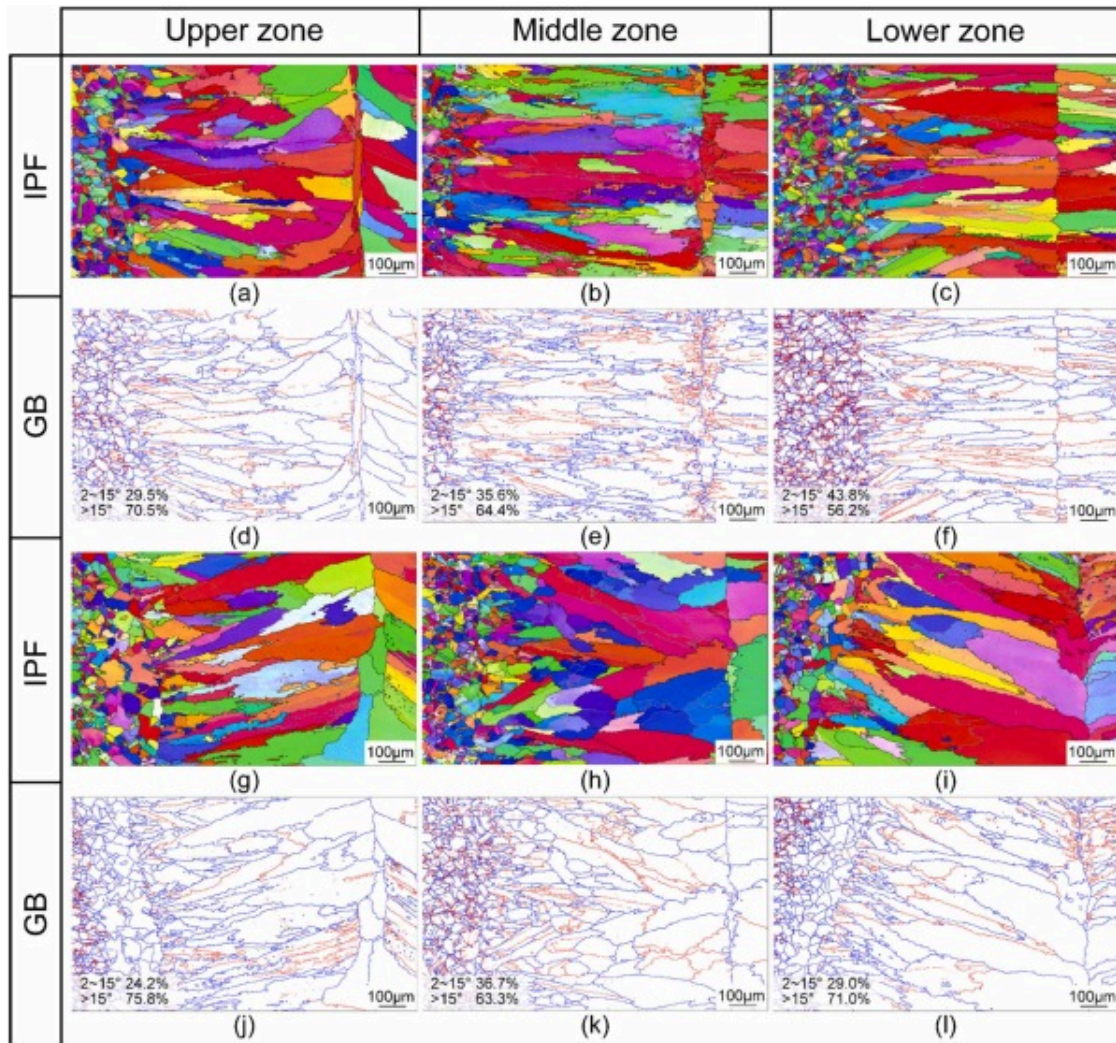
High-speed imaging results indicate that introducing appropriate surface microstructure on the butt joint cross section not only suppresses hump defects but also stabilizes the welding process. The surface microstructure enhances the penetration capability of the laser beam, enabling the transition of the keyhole from a closed state to a fully penetrated state. However, excessive surface roughness of the butt joint cross section is detrimental to weld formation. Guo's simulation results indicate that an excessively large butt joint gap makes it difficult to form a stable molten pool [39]. Based on this, it can be inferred that under parameter set P-4, the relatively large joint gap causes the most concentrated portion of the laser beam energy to act directly on the surface microstructure within the

gap, rapidly melting them to form a molten pool. As shown in the metallographic cross section in Fig. 6c, the fusion zone is insufficiently filled only near the upper and backside surfaces, while the internal region of the plate is continuous and uniform. This indicates that a continuous molten pool is formed during welding with P-4. However, due to insufficient filler metal between the gaps and the excessive detachment of molten material from the pool, severe underfill defects appear on both the upper and backside surfaces of the weld.

4.4. Weld microstructure

A comparative analysis is conducted on the grain growth characteristics and sizes at different thicknesses of the weld after hump defects suppression, aiming to reveal the influence of surface microstructure on the evolution of weld microstructure, as shown in Fig. 10. Firstly, the weld microstructures of both the control group P-0 and the suppression group P-2 are primarily composed of columnar grains. This is attributed to the high temperature gradient (G) and rapid cooling rate (R) in the molten pool during ultra-high-power laser welding, resulting in a high G/R ratio. Such conditions favor the growth of columnar grains as the dominant solidification mode [55]. The phenomenon of competitive grain growth in the weld of P-0 is more pronounced, with grains growing from the fusion line toward the weld center being scarcely observed. Subsequently, a comparison of the grain growth directions reveals that the grains in P-0 grow almost horizontally, whereas those in P-2 are oriented at a certain angle to the horizontal plane. Since grain growth follows the direction of the maximum temperature gradient [56], which is nearly perpendicular to the fusion line, the grain orientation is largely governed by the shape of the weld fusion boundary. Vertically grown grains are observed at the weld centerline of the UZ in both P-0 and P-2, indicating that the horizontal temperature gradient in this region is relatively weak, and that the vertical temperature gradient also contributes to grain growth [57]. Finally, it is noted that the fraction of low-angle grain boundaries (LAGBs) at different thicknesses of the P-2 weld is generally comparable to or lower than that of the P-0 weld, particularly in the LZ. LAGBs in welds essentially represent substructures formed by dislocation accumulation [58]. The lower LAGBs

fraction in the P-2 weld may be attributed to its thermal cycle involving a higher peak temperature and longer high-temperature duration, which promotes dislocation rearrangement and the formation of high-angle grain boundaries (HAGBs) [59].



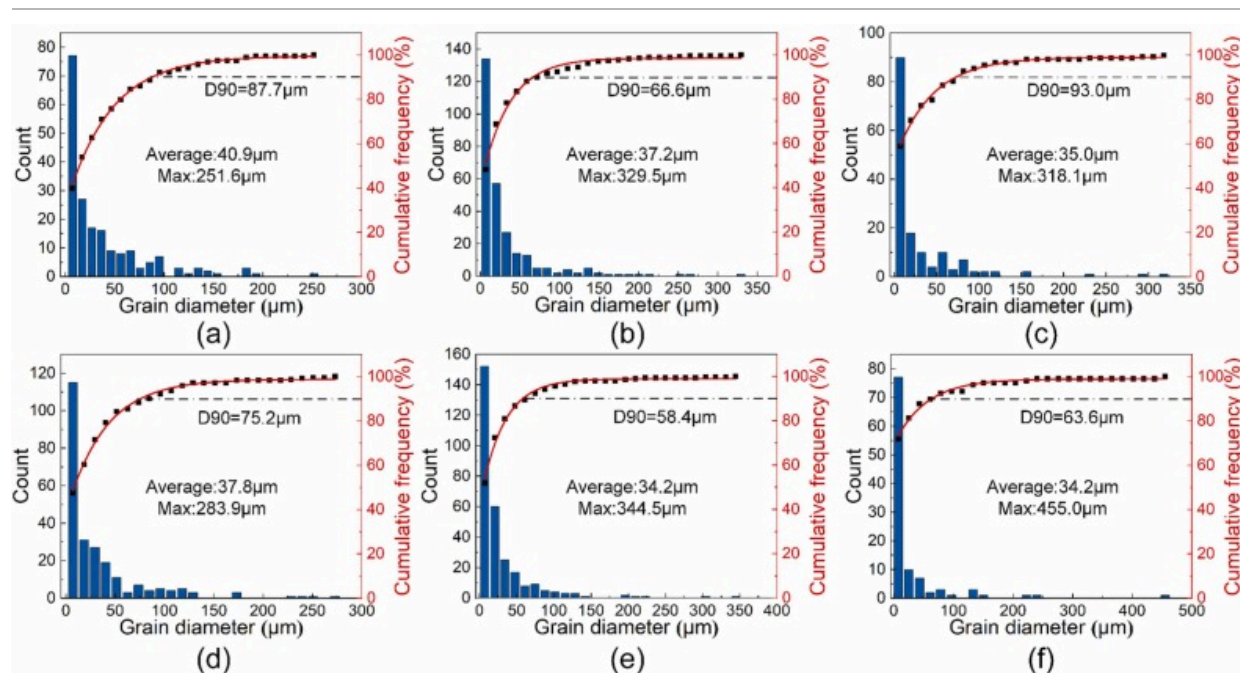
Download: [Download high-res image \(1MB\)](#)

Download: [Download full-size image](#)

Fig. 10. EBSD results at different thicknesses for P-0 and P-2: (a) Inverse pole figure (IPF) mapping of P-0 at UZ; (b) IPF mapping of P-0 at MZ; (c) IPF mapping of P-0 at LZ; (d)

Grain boundary (GB) of P-0 at UZ; (e) GB of P-0 at MZ; (f) GB of P-0 at LZ; (g) IPF mapping of P-2 at UZ; (h) IPF mapping of P-2 at MZ; (i) IPF mapping of P-2 at LZ; (j) GB of P-2 at UZ; (k) GB of P-2 at MZ; (l) GB of P-2 at LZ.

Fig. 11 presents the grain size statistics at different thicknesses of the welds produced with P-0 and P-2 parameters. In the figure, D90 refers to the equivalent circle diameter below which 90 % of the grains fall. For the P-0 weld, the average grain sizes in the UZ, MZ, and LZ are 40.9 μm , 37.2 μm , and 35.0 μm , respectively. At all thicknesses, the P-2 weld exhibits smaller average grain sizes and reduced D90 values compared with the P-0 weld. This indicates that surface microstructuring of the butt joint cross section contributes to grain refinement in the weld under the same welding parameters. Temperature gradient (G) and growth rate (R) jointly determine the morphology and size of grains by influencing undercooling and nucleation rate. The P-2 parameter, with surface microstructuring, absorbs more laser energy, resulting in a higher temperature gradient in the molten pool. Both P-2 and P-0 joint microstructures are columnar grains, but under the same R, the higher G value in P-2 suppresses lateral dendrite growth and increases compositional undercooling, thereby promoting nucleation and refining the weld microstructure.



[Download: Download high-res image \(498KB\)](#)

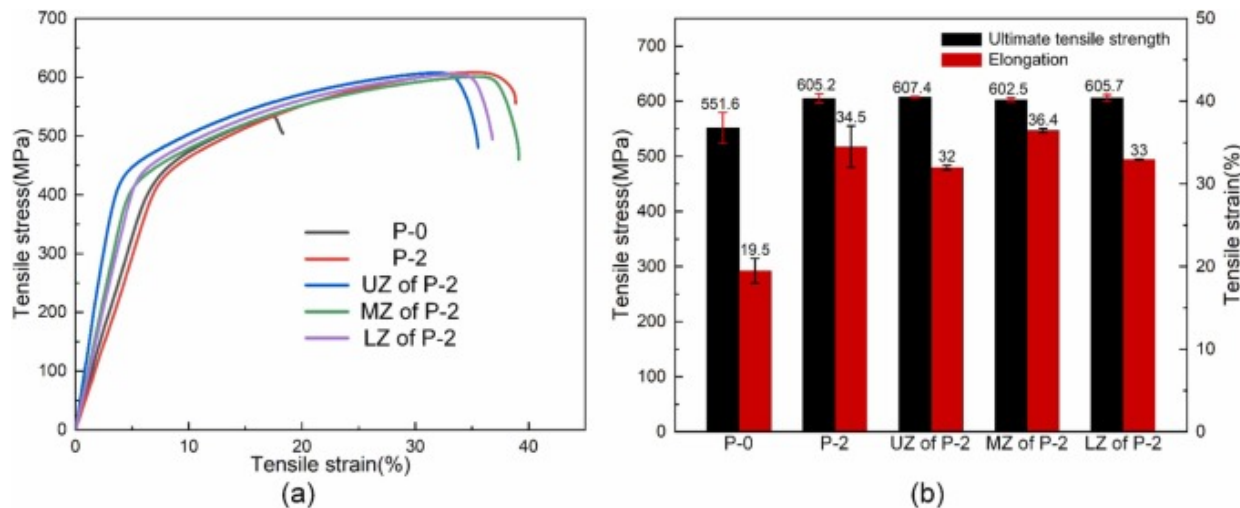
[Download: Download full-size image](#)

Fig. 11. Quantitative analysis of the grain size distribution within the weld seam: (a) Grain size distribution of the weld seam in UZ of P-0; (b) Grain size distribution of the weld seam in MZ of P-0; (c) Grain size distribution of the weld seam in LZ of P-0; (d) Grain size distribution of the weld seam in UZ of P-2; (e) Grain size distribution of the weld seam in MZ of P-2; (f) Grain size distribution of the weld seam in LZ of P-2.

4.5. Mechanical property and fracture morphology

As shown in Fig. 12, the tensile results indicate that the P-0 weld with hump defects exhibits a tensile strength of 87 % of the base metal (630 MPa) and an elongation of only 37 % of the base metal (53 %). Hump defects severely deteriorates the tensile performance of the welded joint [60]. After defects suppression using the P-2 parameter, The tensile strength and elongation reach 96 % and 65 % of those of the base metal, respectively. It demonstrates a significant improvement in joint mechanical performance. This

enhancement is mainly attributed to the improved weld formation quality. The tensile strengths at different thicknesses are all above 95 % of the base metal, with negligible variation among them. The elongation values at different thicknesses also show minimal differences, with the MZ slightly higher than the UZ and LZ.



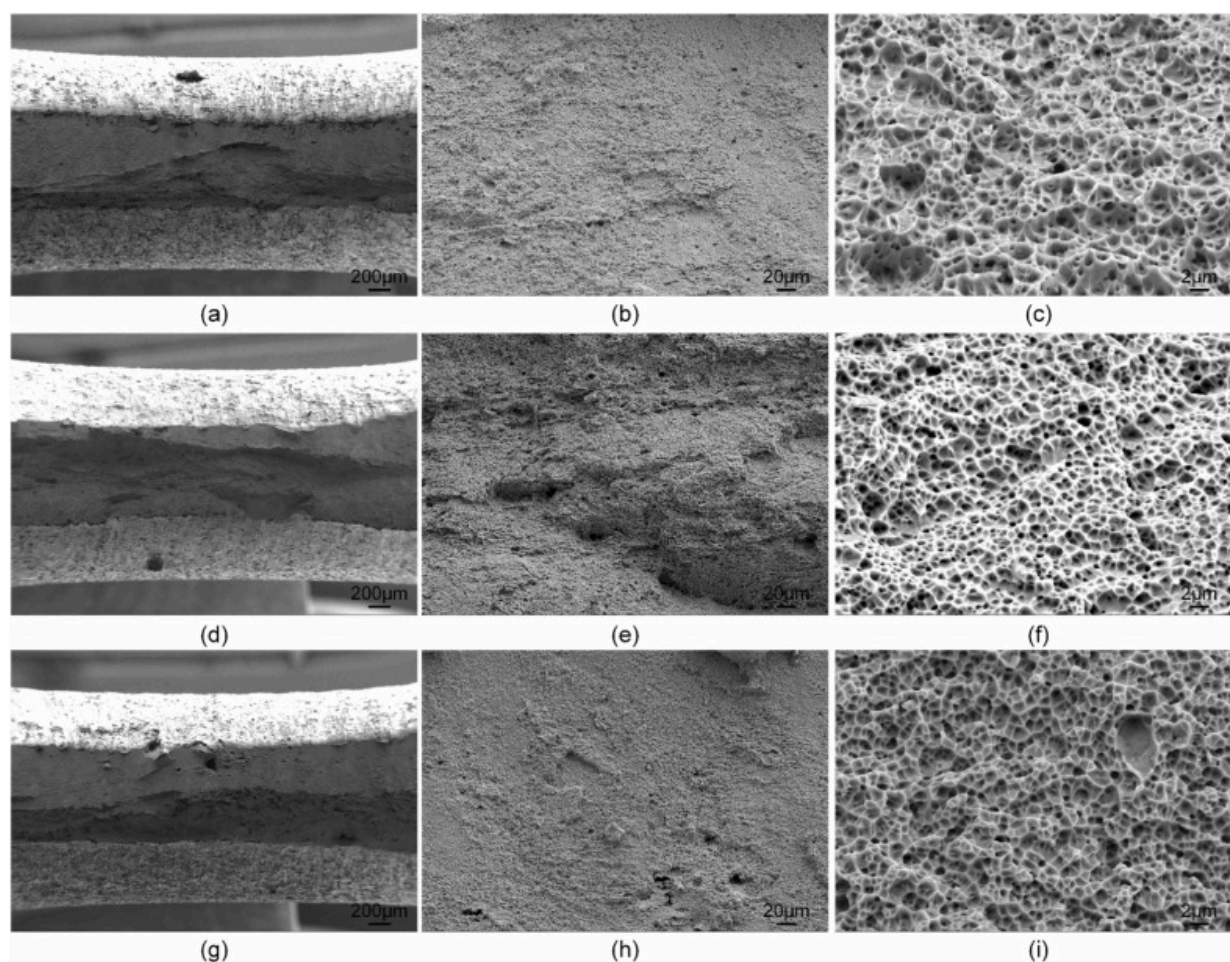
[Download: Download high-res image \(195KB\)](#)

[Download: Download full-size image](#)

Fig. 12. Tensile curves and bar charts: (a) Tensile strength curves of the full-thickness welds for P-0 and P-2, as well as the layered tensile curves for P-2; (b) Bar charts of the tensile strength and elongation for the full-thickness welds with P-0 and P-2 parameters, along with the layered tensile strength and elongation results for P-2. Error bars indicated a 95 % confidence interval (CI).

Fracture morphologies of the tensile specimens at different thicknesses under the P-2 condition were examined. All three sections exhibit noticeable necking, indicating good ductility of the welded joint [61]. High-magnification SEM images reveal typical dimples, indicating that the fracture mode is ductile. As shown in Fig. 13e, the fracture surface of the MZ is less flat than those in Fig. 13b,h, reflecting a more tortuous crack propagation path. Compared with the UZ and LZ, the MZ exhibits deeper and denser dimples with a

more uniform distribution, suggesting more extensive dislocation slip prior to fracture and a well-developed dimple growth process. This observation is consistent with the higher elongation measured in the MZ.



[Download: Download high-res image \(1MB\)](#)

[Download: Download full-size image](#)

Fig. 13. Fracture morphologies: (a-c) Fracture morphology of the UZ in P-2; (d-f) Fracture morphology of the MZ in P-2; (g-i) Fracture morphology of the LZ in P-2.

5. Discussions

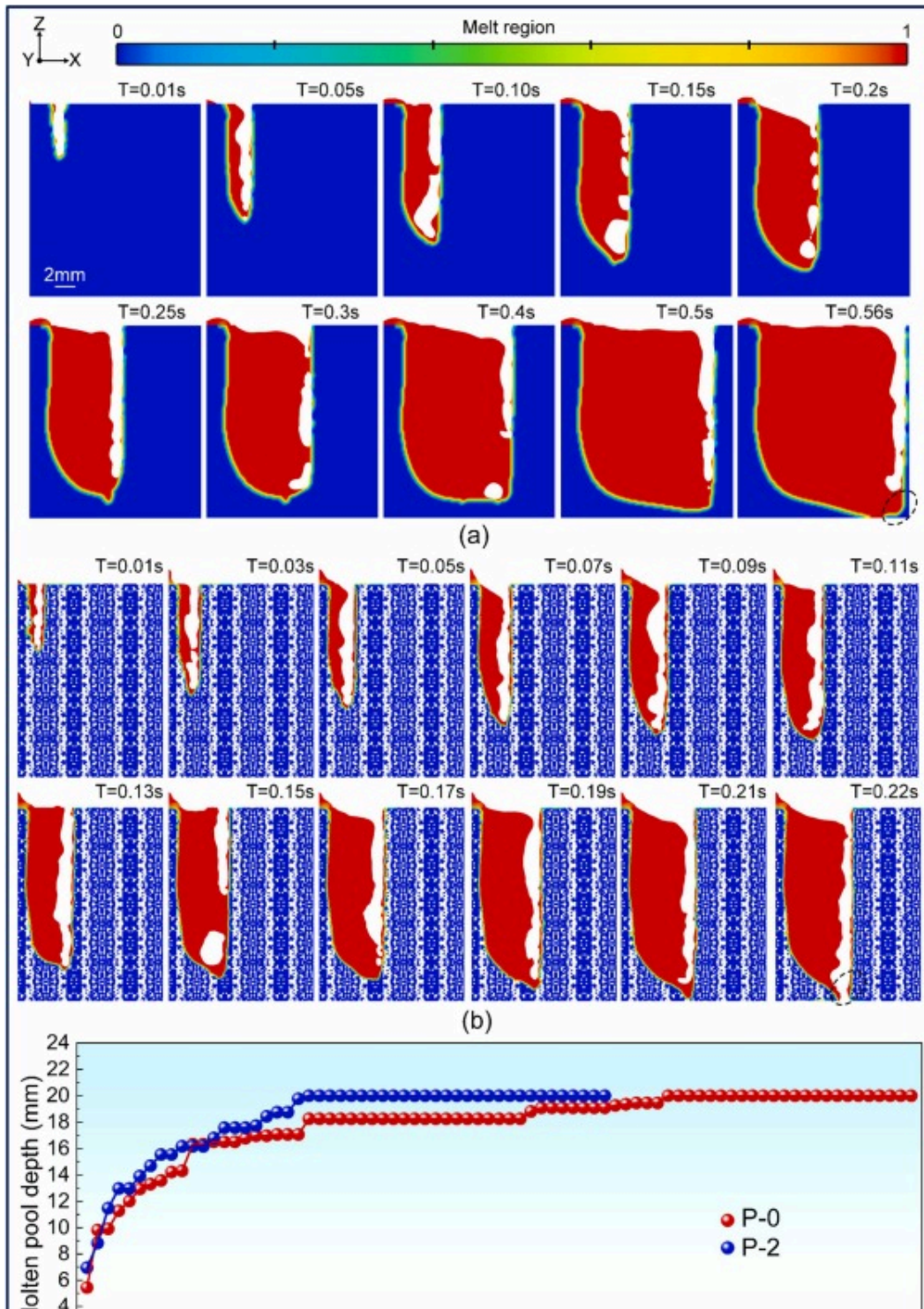
5.1. The direct effect of microstructure on the welding process

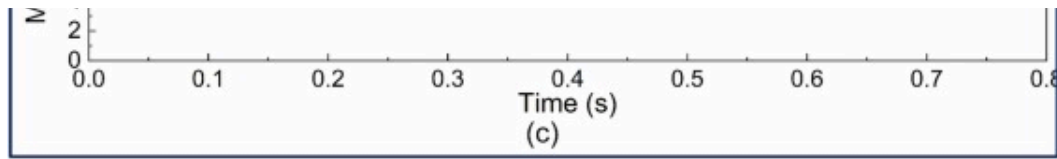
A CFD model for ultra-high-power laser welding of thick plates is developed, taking into account the surface microstructural morphology of the butt joint cross section. Furthermore, the weld widths at the upper, middle, and bottom sections of the cross-section, as well as the weld areas, were measured for both the experimental and simulation results, with the data presented in Table 4. The comparison indicates that the model exhibits overall high accuracy. The relatively lower accuracy in the upper section of the weld may be due to the model not accounting for the plasma generated during the welding process. In ultra-high-power laser welding, the formation of plasma is more pronounced, and its stronger scattering and refraction effects on the laser beam can result in wider weld widths in the experimental results compared to the simulation.

Table 4. Comparison of weld seam width between experimental and simulated results.

	Parameter P-0			Parameter P-2		
	Experiment	Simulation	Precision	Experiment	Simulation	Precision
Weld seam upper width (W_u mm)	2.8	1.6	57 %	3.3	1.8	55 %
Weld seam middle width (W_m mm)	1.9	2.2	86 %	1.9	2.3	83 %
Weld seam bottom width (W_b mm)	1.8	2.3	78 %	2.7	2.6	96 %
Weld area (mm^2)	35.6	41.5	86 %	40.9	45.2	90 %

As shown in Fig. 2, the simulated cross-sectional profile agrees well with the experimental metallographic cross section, demonstrating good model accuracy. Fig. 14 shows the evolution of the melt region from the initial moment to the stage just reaching full penetration. The cross-sectional view corresponds to the X-Z longitudinal section at the center along the Y-axis. In the figure, the blue area represents the unmelted region, while the red area indicates the region melted by laser irradiation. Since the butt joint cross section under the P-0 condition is machined and the two plates are nearly in full contact during assembly, the numerical model for P-0 is established in the form of flat-plate surfacing, similar to other studies [62], [63]. Therefore, the cross section shown in Fig. 14a is flat and structurally intact. In contrast, the numerical model for parameter P-2 incorporates the surface microstructure of the butt joint cross section, and in Fig. 14b, only the blue region represents the base material defined by the surface microstructure.





Download: [Download high-res image \(1MB\)](#)

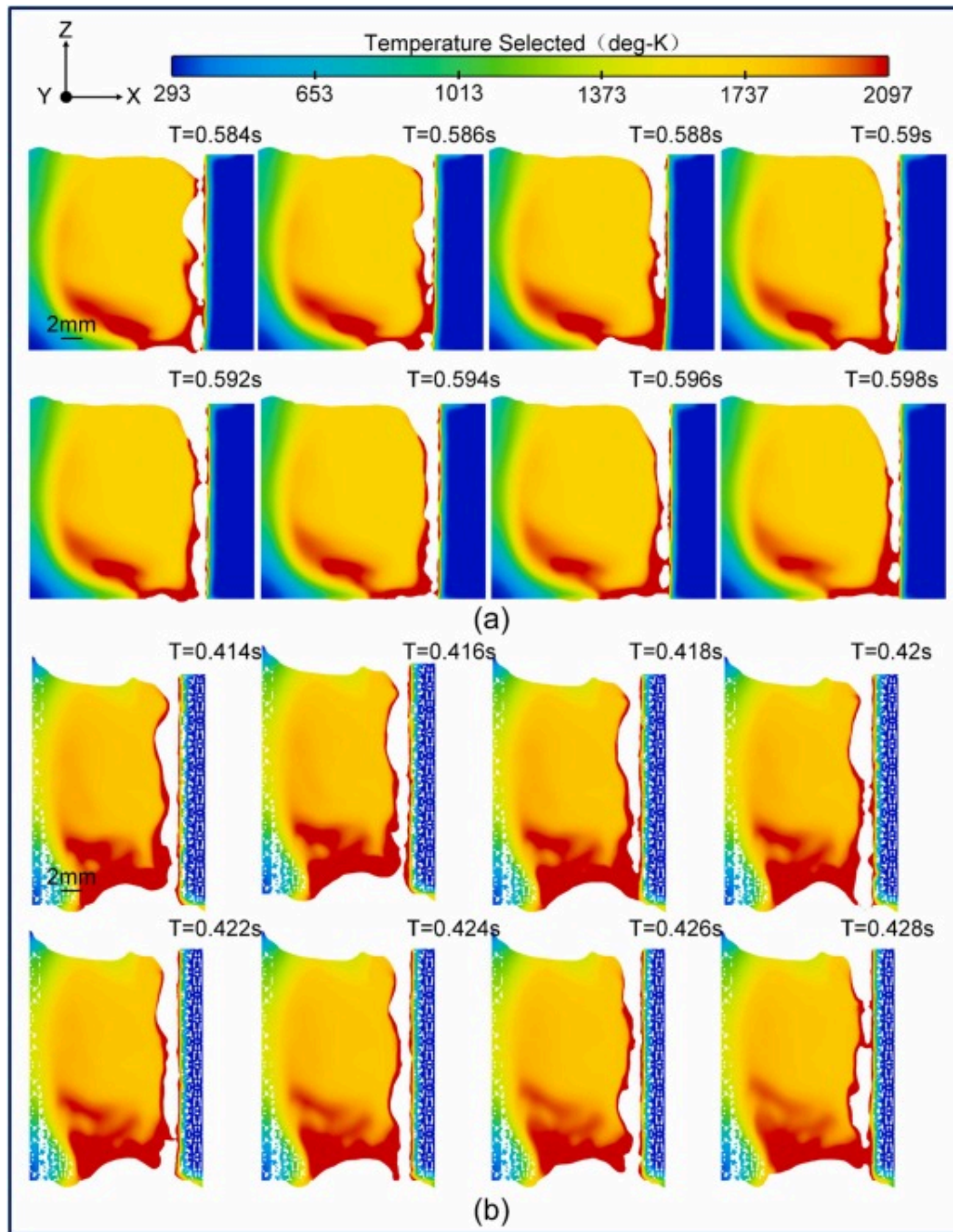
Download: [Download full-size image](#)

Fig. 14. Numerical simulation results of the molten zone during the incomplete penetration stage: (a) Morphological evolution of the melt zone cross-section for parameter P-0, which exhibits hump defects; (b) Morphological evolution of the melt zone cross-section for parameter P-2, which suppresses hump defects; (c) Curve of maximum molten zone depth variation.

The evolution of the melt region morphology shows that the molten pool under parameter P-2 achieves full penetration through the plate in the Z-direction at 0.22 s, which is significantly shorter than the 0.56 s under parameter P-0. Multiplying the penetration time of P-0 with the welding speed of 25 mm/s, the unpenetrated length along the X-direction on the bottom surface of the weld is estimated to be approximately 14 mm. This distance closely matches the experimental results shown in Fig. 6a, and the consistency between simulation and experiment further validates the reliability of the numerical model. Additionally, it can be observed that the bottom of the front wall of the keyhole under parameter P-0 is curved, indicating poor laser beam penetration capability. At the moment of molten pool penetration, the keyhole does not simultaneously achieve full penetration. Preliminary process experiments [38] indicate that parameter P-0 has insufficient heat input and is at the critical state between full penetration and incomplete penetration. If welding speed is increased slightly to reduce heat input, incomplete penetration will occur. In contrast, the front wall of the keyhole under parameter P-2 remains almost vertical, demonstrating strong laser beam penetration capability, and the keyhole achieves full penetration simultaneously with the molten pool.

Data on the maximum molten pool depth at 0.01 s intervals are collected, and the curve is shown in [Fig. 14c](#). The curve indicates that at every time point, the maximum molten pool depth for parameter P-2 is greater than that for parameter P-0. After preparing the surface microstructure on the butt joint, the maximum depth of the molten zone steadily increases over time, reflecting the stability of the welding process. For parameter P-0, the maximum depth of the molten zone suddenly increases at around 0.13 s and 0.22 s, then remains unchanged for a period of time afterward. This rapid downward expansion of the molten pool indicates instability in the welding process.

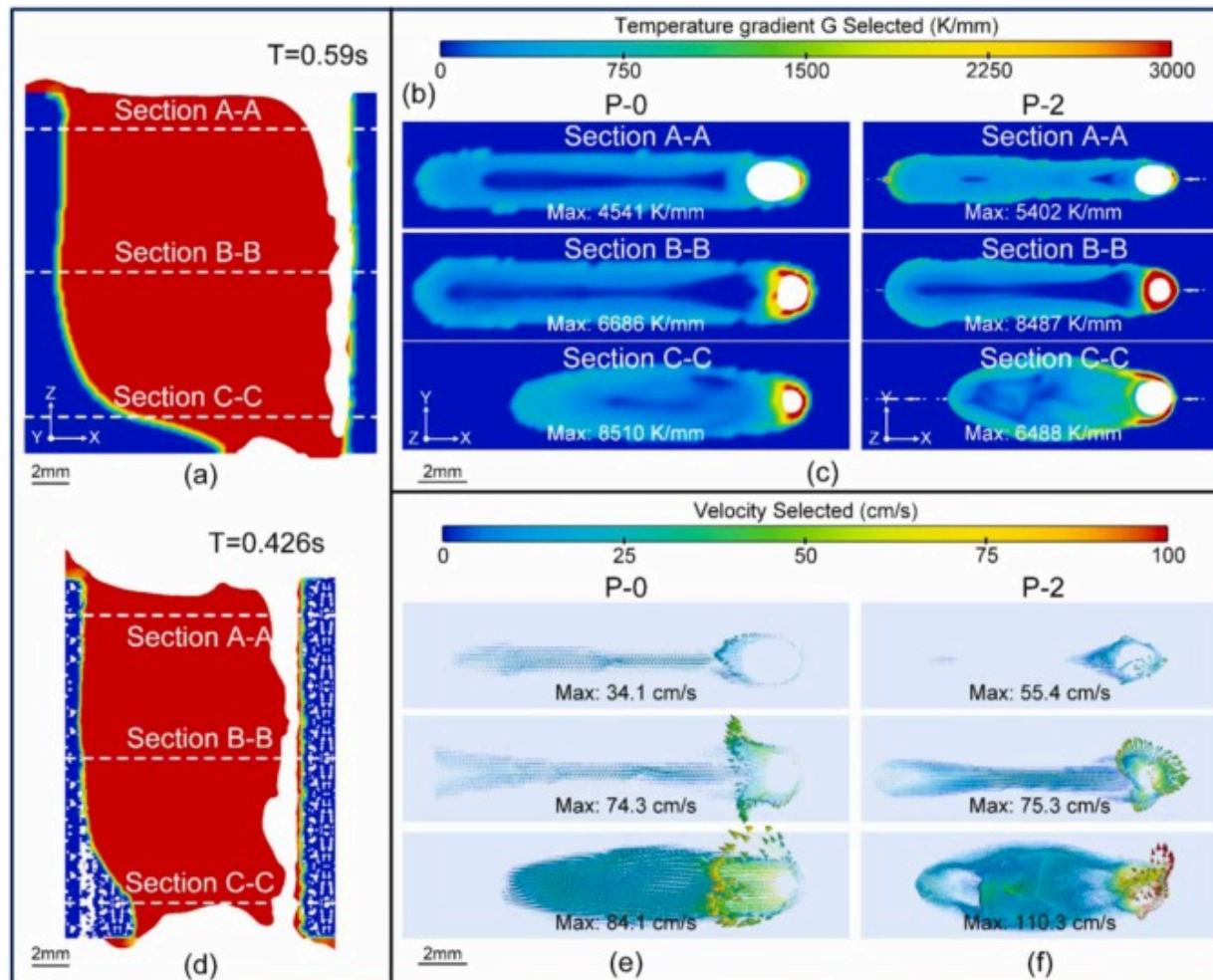
The high-speed camera images presented in [Section 4.3](#) are captured at intervals of 220 ms. To analyze the keyhole status and its evolution over a continuous time period, [Fig. 15](#) displays temperature field results during the molten pool penetration stage with a finer time interval of 2 ms. The red regions in the figure represent molten pool areas with temperatures 400 K above the liquidus temperature of 1697 K. Within the 14 ms period, the keyhole of parameter P-0 remains mostly closed, prone to collapse at the bottom, with small high-temperature zones at the molten pool base. In contrast, the keyhole for parameter P-2 stays nearly fully penetrated; even when collapse occurs, it quickly recovers to the penetrated state within the 2 ms intervals, and there are larger high-temperature zones at the molten pool bottom. ([Fig. 16](#), [Fig. 17](#))



Download: [Download high-res image \(675KB\)](#)

[Download: Download full-size image](#)

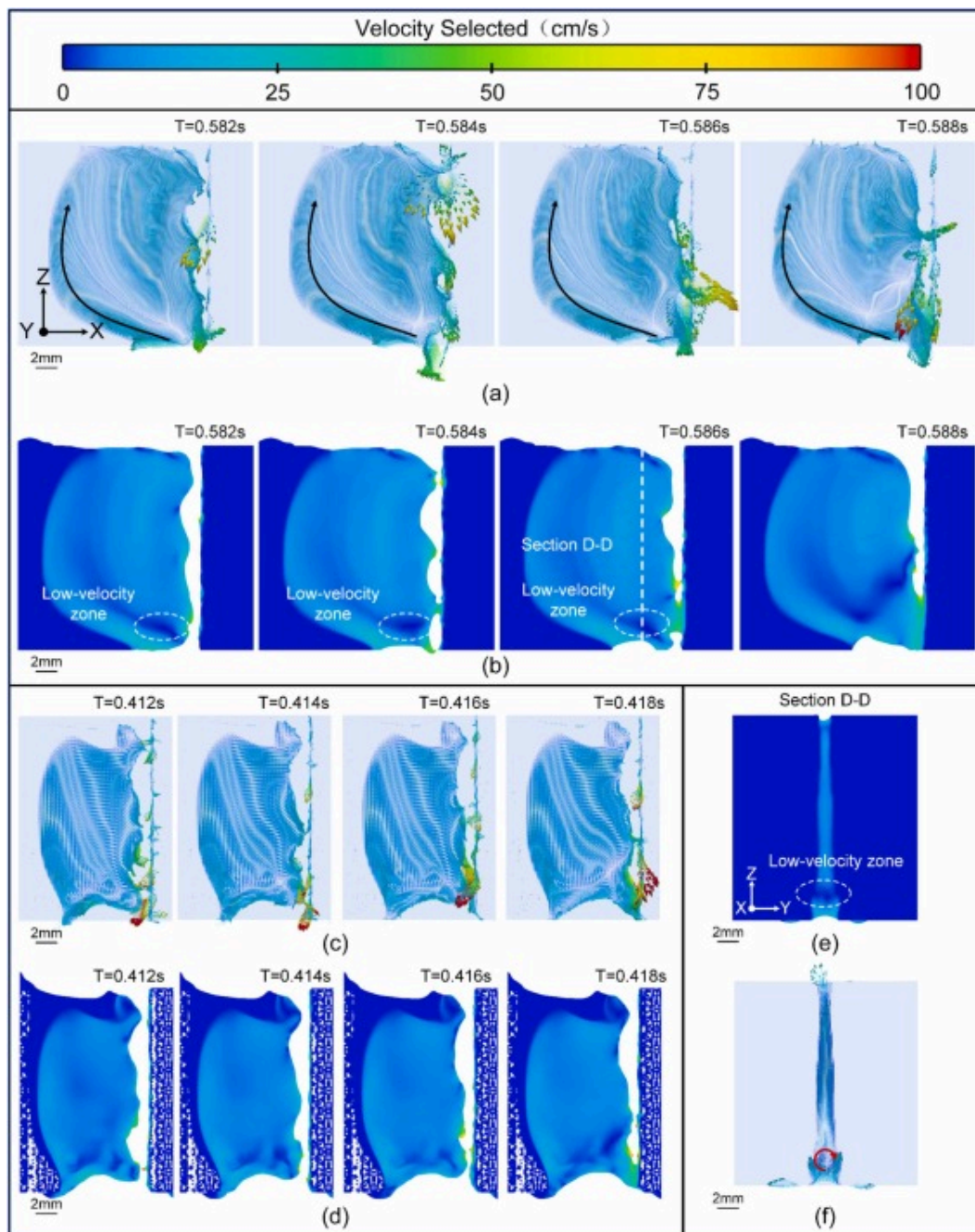
Fig. 15. Numerical simulation results of the molten pool temperature field during the penetration stage: (a) Parameter P-0 with hump defects occurrence; (b) Parameter P-2 with hump defects suppression.



[Download: Download high-res image \(548KB\)](#)

[Download: Download full-size image](#)

Fig. 16. Comparison of heat and mass transfer behaviors at different thicknesses: (a) Melt zone and keyhole morphology in the longitudinal section for parameter P-0; (b) Temperature gradient fields on the X-Y sections at 2 mm, 10 mm, and 18 mm below the top surface for parameter P-0; (c) Temperature gradient fields on the X-Y sections at 2 mm, 10 mm, and 18 mm below the top surface for parameter P-2; (d) Melt zone and keyhole morphology in the longitudinal section for parameter P-2; (e) Velocity fields around the keyhole on the X-Y sections at 2 mm, 10 mm, and 18 mm below the top surface for parameter P-0; (f) Velocity fields around the keyhole on the X-Y sections at 2 mm, 10 mm, and 18 mm below the top surface for parameter P-2.

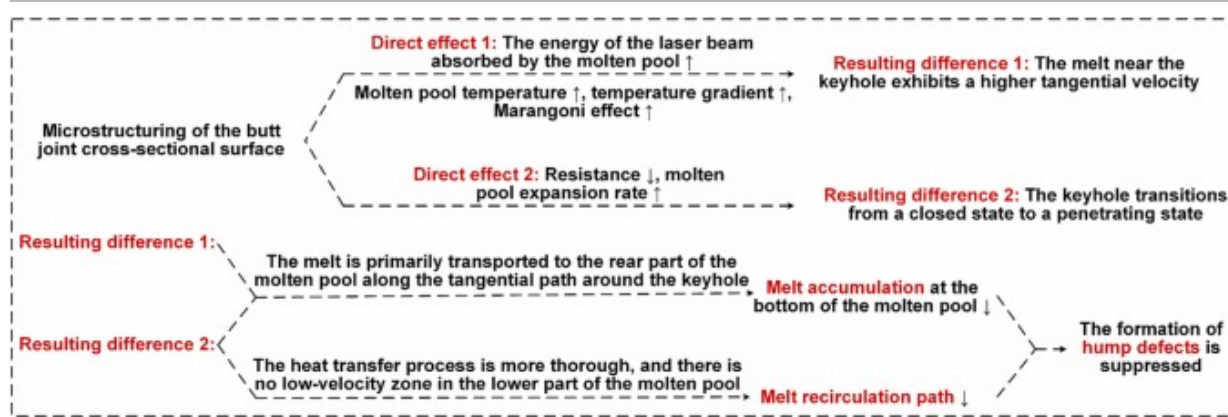


Download: [Download high-res image \(823KB\)](#)

Download: [Download full-size image](#)

Fig. 17. Simulated results of temperature and velocity vector fields on the longitudinal sections for parameters P-0 and P-2: (a) Velocity vector field on the longitudinal section for parameter P-0; (b) Velocity field on the longitudinal section for parameter P-0; (c) Velocity vector field on the longitudinal section for parameter P-2; (d) Velocity field on the longitudinal section for parameter P-2; (e) Velocity field on the transverse section of the low-velocity region for parameter P-0 at $T = 0.586$ s; (f) Velocity vector field on the transverse section of the low-velocity region for parameter P-0 at $T = 0.586$ s.

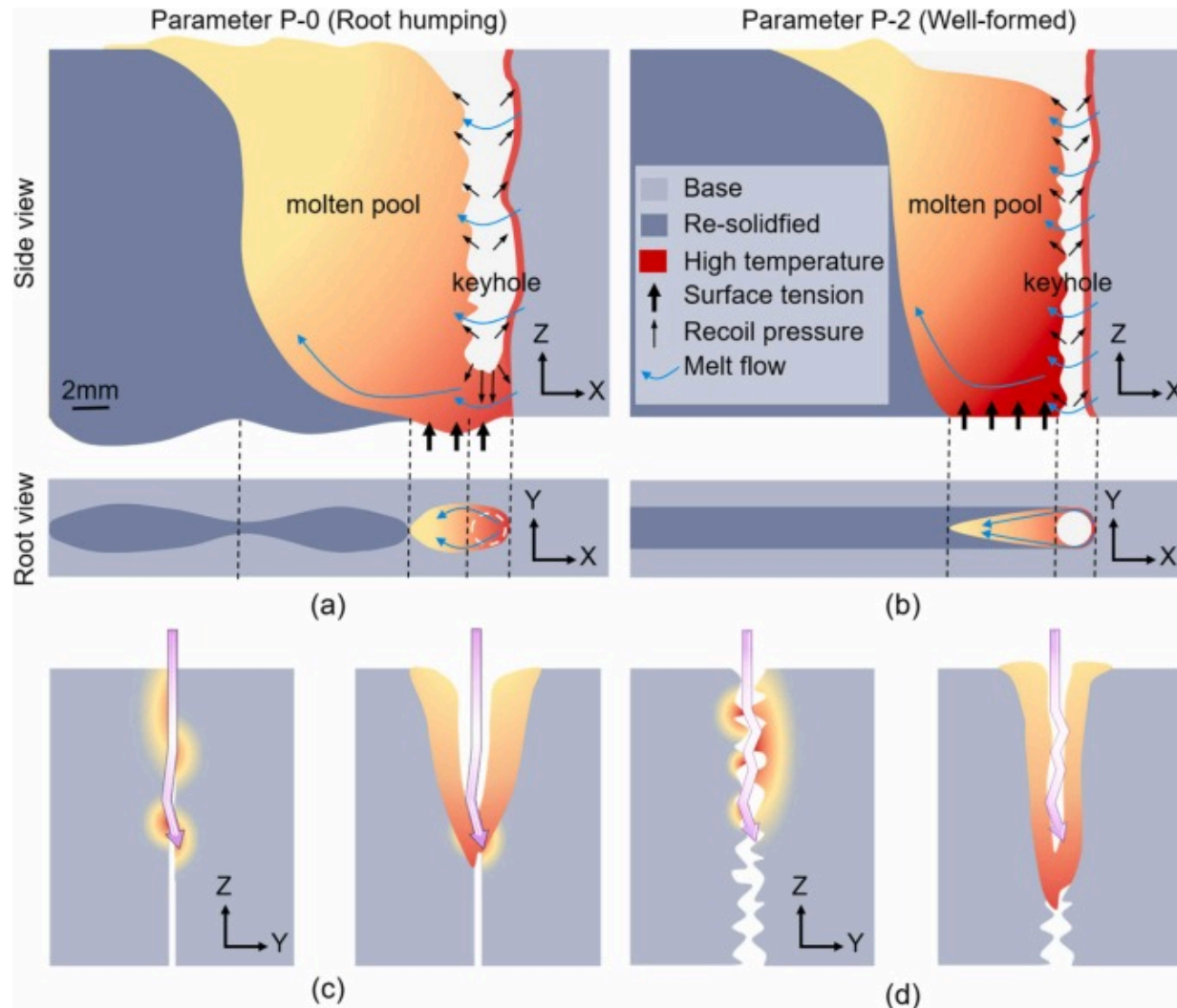
As shown in Fig. 18, the microstructure of the butt joint cross-section has two direct effects on the welding process. The first is an increase in the absorption of laser energy. As illustrated in Fig. 19d, when the laser beam travels along the welding direction, it can directly enter the butt joint gap formed by the surface microstructure. After undergoing refraction and absorption by the diversified surface microstructures, the molten material forms a molten pool. Once the molten pool and keyhole are formed, the laser beam acts on the keyhole walls and can no longer directly interact with the surface microstructure. Energy transfer is then dominated by the multiple reflection effect within the keyhole [64]. Although the refraction and absorption of the beam by the aforementioned microstructures are brief and occur mainly in the upper part of the plate, the laser beam continuously interacts with the unmelted material ahead as it travels along the welding direction. The multiple reflection and absorption processes induced by the cross-sectional microstructure persist, enhancing the molten pool's absorption of the laser beam energy [65], [66]. The statistics of weld width and molten area in Fig. 7 verify the above analysis. After surface microstructuring, the weld upper surface width increased by 17.0 %, and the weld area increased by 14.9 %. The second direct effect of the butt joint cross-sectional microstructure on the welding process is that, after surface microstructuring, the butt joint has a gap of approximately 40 μm . This reduces the amount of material that needs to be melted, which clearly facilitates the downward expansion and forward propagation of the molten pool and keyhole.



Download: [Download high-res image \(256KB\)](#)

Download: [Download full-size image](#)

Fig. 18. Mechanism of hump defects suppression and the effect of surface microstructure at the butt joint interface.



[Download: Download high-res image \(284KB\)](#)

[Download: Download full-size image](#)

Fig. 19. Suppression mechanism of hump defects by microstructures: (a) Molten pool characteristics and keyhole state for parameter P-0, where hump defects occur; (b) Molten pool characteristics and keyhole state for parameter P-2, where hump defects are suppressed; (c) Interaction of the laser beam with the milled cross-section and the initial

molten pool formation process; (d) Interaction of the laser beam with the microstructured cross-section and the initial molten pool formation process.

High-speed imaging and simulation results indicate that the microstructuring of the butt joint cross section facilitates the expansion of the molten pool and promotes the transformation of the keyhole morphology from a closed to a fully penetrated state during welding. Eq. (12) reflects the positive correlation between recoil pressure and temperature. On one hand, microstructuring increases the energy absorbed by the molten pool, raising its temperature and generating greater recoil pressure to drive the downward expansion of the keyhole. On the other hand, the microstructure presents less resistance to keyhole formation and expansion, leading to a transformation in the keyhole morphology.

5.2. Mechanism for suppressing hump defects

During laser deep penetration welding, a keyhole is formed. As the laser beam travels along the welding direction, unmelted material in front of the keyhole melts and flows into the molten pool behind the keyhole, where it eventually solidifies to form the weld. This section analyzes the mechanism of humps suppression from the perspectives of melt transport and solidification.

As discussed earlier, the keyhole under the P-0 parameter remains mostly in a non-penetrating state during welding. As shown in Fig. 16a, molten metal in front of the keyhole flows to the rear of the molten pool through the keyhole bottom. In contrast, the keyhole under the P-2 parameter remains largely in a penetrating state, blocking this transport path. An alternative path for the molten metal in front of the keyhole to reach the rear molten pool is via circumferential flow along the keyhole wall [67], [68]. To compare the intensity of circumferential flow under the two parameters, X-Y cross-sections are extracted at depths of 2 mm, 10 mm, and 18 mm from the upper surface. The temperature gradient and velocity field around the keyhole are then analyzed, as shown in Fig. 16. By comparing the velocity fields around the keyhole in Fig. 16e, it is evident

that the circumferential flow velocity around the keyhole at all depths is higher under parameter P-2 than under P-0. This indicates that molten metal transport in P-2 is primarily driven by circumferential flow around the keyhole, whereas in P-0, a significantly larger amount of molten metal is delivered from the keyhole bottom to the rear part of the molten pool. This provides the material basis for the formation of hump defects on the backside of the weld. As shown in [Fig. 16b](#), the temperature gradient around the keyhole is consistently higher under parameter P-2 than under P-0 at all depths. This can be attributed to the enhanced absorption of laser energy by the surface microstructure at the joint interface, which increases the melt temperature around the keyhole wall and results in a steeper temperature gradient. Since the temperature gradient is the primary driving force of the Marangoni effect [\[69\]](#), the molten metal near the keyhole under parameter P-2 exhibits higher circumferential velocity.

As shown in [Fig. 17](#), the longitudinal sections of parameters P-0 and P-2 are extracted, along with their corresponding velocity field and velocity vector field. In the velocity field, the blue regions indicate areas where the molten pool flow velocity is close to 0 cm/s, while in the velocity vector field, red arrows represent locations where the flow velocity exceeds 100 cm/s.

As shown in [Fig. 17b](#), a localized low-velocity zone is observed in the lower part of the molten pool, which hinders the inward contraction of the molten metal at the bottom. A transverse section at the location of this low-velocity zone is extracted, and its velocity field and velocity vector field are shown in [Fig. 17e,f](#), respectively, clearly revealing a velocity discontinuity caused by the low-velocity region. This phenomenon may be attributed to the dominant clockwise flow pattern inside the molten pool of parameter P-0, as indicated by the black arrows in [Fig. 17a](#). The thermal transport of the molten metal along the flow path results in higher temperatures, as shown in [Fig. 15a](#). In contrast, the low-velocity zone is located in a relatively cooler zone, where the increased local viscosity in the mushy region leads to the formation of a low-velocity zone [\[70\]](#).

Based on the preceding analysis, for parameter P-0, the molten material in front of the keyhole tends to flow into the lower part of the molten pool after melting, which provides the material basis for defects formation. However, due to the restricted transport path, this molten metal is hindered from contracting inward, making it more prone to form hump defects. In contrast, for parameter P-2, the molten metal around the keyhole exhibits higher circumferential velocity, and the keyhole remains in a fully penetrated state, as shown in Fig. 19b. The molten metal is mainly transported to the rear of the molten pool along the circumferential path of the keyhole, resulting in less accumulation at the bottom of the molten pool and thus suppressing the hump defects. Following the approach of other researchers [71], Fig. 18 presents a topological schematic illustrating the effect of the butt joint surface microstructure and the mechanism of hump defects suppression.

6. Conclusions

In this study, a microstructuring method was applied to the butt joint surfaces, successfully suppressing hump defects in single-pass welding of 20 mm thick stainless steel plates. A ultra-high-power thick plate welding numerical model incorporating the surface microstructure morphology of the butt joint was established. The suppression mechanism of hump defects by surface microstructuring was revealed. The following conclusions can be drawn:

1. Preparing microstructures with appropriate roughness on the butt joint cross-section facilitates hump suppression under identical welding parameters, enabling good single-pass formation of 20 mm-thick stainless steel. In this study, the butt gap formed by suitable microstructures is approximately $1/15$ of the laser spot diameter and do not exceed $2/15$.
2. The surface microstructuring of the butt joint promotes molten pool expansion and facilitates the transition of the keyhole from a closed state to a fully penetrated state during welding. It also enhances the molten pool's

absorption of laser energy, resulting in a higher temperature gradient near the keyhole walls. Driven by the Marangoni effect, the molten metal near the keyhole exhibits increased circumferential velocity, enabling more effective melt transport toward the rear of the molten pool.

3. The above physical mechanisms are independent of material composition and apply to all welding processes dominated by liquid-metal flow in theory. However, the universality of these mechanisms has certain limitations. For high-thermal-conductivity materials such as aluminum and copper, rapid heat dissipation in the molten pool weakens the thermal effects induced by microstructures. For high-viscosity zinc alloys, the Marangoni effect is relatively weak, resulting in limited flow acceleration. The universality of the proposed validation method requires further experimental verification in future studies.
4. For the parameters associated with hump defect formation, the keyhole remains mostly in a closed state. The molten metal surrounding the keyhole exhibits lower circumferential velocity, making it easier for the melted material at the leading edge to flow toward the rear of the molten pool through the bottom of the keyhole. Moreover, the presence of a low-temperature and low-velocity region in the lower part of the molten pool hinders the inward shrinkage of the molten metal, making the accumulated melt more likely to solidify and form hump defects.
5. The weld treated with surface microstructuring exhibits a lower fraction of low-angle grain boundaries, which is attributed to the higher peak temperature and longer high-temperature dwell time that promote dislocation rearrangement into high-angle grain boundaries. It also features finer grain sizes, resulting from the rapid cooling during the solidification process. After surface microstructuring, the joint's overall tensile strength increases to 96 % and the elongation to 65 % of the base material. The

improvement in mechanical properties is mainly attributed to the enhanced weld formation quality.

6. Compared with existing hump-suppression methods such as the vacuum method, magnetic field assistance, and gas-flow assistance, the surface microstructuring approach proposed in this study offers the advantages of requiring no additional auxiliary equipment, high flexibility, and strong universality. In theory, it can be applied to the joining of long and curved welds of various thicknesses. Its limitations lie in the fact that surface microstructuring introduces an additional processing step, and an effective correlation between different materials, different welding systems, and suitable surface roughness has not yet been established. Further research is needed to address these shortcomings, improve the efficiency of the microstructuring process, and develop rapid methods for obtaining suitable microstructure roughness.

CRediT authorship contribution statement

Chunming Wang: Writing – review & editing, Supervision, Conceptualization. **Zehui Liu:** Methodology, Formal analysis, Data curation. **Zhongshun Zhao:** Formal analysis, Data curation. **Yiyang Hu:** Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Fei Yan:** Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was mainly supported by the National Natural Science Foundation of China (Grant No. [52375377](#)). In addition, we would also like to express our gratitude to the Analysis and Test Center of Huazhong University of Science and Technology (HUST) and the State Key Laboratory of Digital Manufacturing Equipment & Technology of HUST for their friendly cooperation.

References

- [1] H. Yang, J. Chen, X. Zhao, N. Huda, A.R.H. Midawi, A.P. Gerlich
Effect of laser beam wobbling and welding speed in X80 steel wire-fed laser root welds
J. Mater. Process. Technol., 327 (2024), [10.1016/j.jmatprotec.2024.118383](#) ↗
[Google Scholar](#) ↗
- [2] E. de Sonis, S. Depinoy, P.-F. Giroux, H. Maskrot, P. Wident, O. Hercher, F. Villaret, A.-F. Gourgues-Lorenzon
Microstructure-Toughness relationships in 316L stainless steel produced by laser powder bed fusion
Mater. Sci. Eng. aStruct. Mater. Prop. Microstruct. Process., 877 (2023), [10.1016/j.msea.2023.145179](#) ↗
[Google Scholar](#) ↗
- [3] S.L. Mannan, S.C. Chetal, B. Raj, S.B. Bhoje
Selection of materials for prototype fast breeder reactor
Trans. Indian Inst. Met., 56 (2) (2003), pp. 155-178
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [4] P. Vasantharaja, M. Vasudevan, V. Maduraimuthu
Effect of arc welding processes on the weld attributes of type 316LN stainless steel weld joint
Trans. Indian Inst. Met., 71 (1) (2018), pp. 127-137, [10.1007/s12666-017-1162-2](#) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [5] T.S. Kumar, M.K. Dash, A. Nagesha
Deformation and damage assessment in type 316 LN stainless steel weld joint under isothermal and thermomechanical cyclic loading
Mater. Sci. Eng. aStruct. Mater. Prop. Microstruct. Process., 849 (2022),
[10.1016/j.msea.2022.143494](https://doi.org/10.1016/j.msea.2022.143494) ↗
[Google Scholar ↗](#)
- [6] M. Sabzi, S.H.M. Anijdan, A.R.B. Chalandar, N. Park, H.R. Jafarian, A.R. Eivani
An experimental investigation on the effect of gas tungsten arc welding current modes upon the microstructure, mechanical, and fractography properties of welded joints of two grades of AISI 316L and AISI310S alloy metal sheets
Mater. Sci. Eng. aStruct. Mater. Prop. Microstruct. Process., 840 (2022),
[10.1016/j.msea.2022.142877](https://doi.org/10.1016/j.msea.2022.142877) ↗
[Google Scholar ↗](#)
- [7] Y. Pan, Q. Zhu, Y. Wang, L. Jiang, Y. Yan, Z. Li, J. Wang, Y. Zhu, C. Li
Single-pass ultra-high-power laser welding of thick nickel-based GH3535 superalloy: microstructure evolution and microhardness
Mater. Lett., 376 (2024), [10.1016/j.matlet.2024.137239](https://doi.org/10.1016/j.matlet.2024.137239) ↗
[Google Scholar ↗](#)
- [8] Y. Wan, W. Jiang, W. Wei, X. Xie, M. Song, G. Xu, X. Xie, X. Zhai
Characterization of inhomogeneous microstructure and mechanical property in an ultra-thick duplex stainless steel welding joint
Mater. Sci. Eng. aStruct. Mater. Prop. Microstruct. Process., 822 (2021),
[10.1016/j.msea.2021.141640](https://doi.org/10.1016/j.msea.2021.141640) ↗
[Google Scholar ↗](#)

- [9] S. Reich, M. Goesmann, M. Lueck, J. Osterholz, Laser penetration of metal targets with high powers of up to 120 kW, Conference on High Power Lasers - Technology and Systems, Platforms, Effects VI, Amsterdam, NETHERLANDS, 2023.
[Google Scholar ↗](#)
- [10] Y. Li, S. Geng, L. Shu, Y. Li, P. Jiang
Ultra-high-power laser welding of thick-section steel: current research progress and future perspectives
Opt. Laser Technol., 167 (2023), [10.1016/j.optlastec.2023.109663 ↗](#)
[Google Scholar ↗](#)
- [11] J. Karlsson, P. Norman, A.F.H. Kaplan, P. Rubin, J. Lamas, A. Yanez
Observation of the mechanisms causing two kinds of undercut during laser hybrid arc welding
Appl. Surf. Sci., 257 (17) (2011), pp. 7501-7506, [10.1016/j.apsusc.2011.03.068 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [12] J. Powell, T. Ilar, J. Frostevarg, M.J. Torkamany, S.-J. Na, D. Petring, L. Zhang, A.F.H. Kaplan
Weld root instabilities in fiber laser welding
J. Laser Appl., 27 (2015), [10.2351/1.4906390 ↗](#)
[Google Scholar ↗](#)
- [13] S. Katayama, A. Yohei, M. Mizutani, Y. Kawahito, Development of Deep Penetration Welding Technology with High Brightness Laser under Vacuum, 6th International WLT Conference on Lasers in Manufacturing (LiM), Munich, GERMANY, 2011, pp. 75-80.
[Google Scholar ↗](#)
- [14] A. Youhei, K. Yousuke, N. Hiroshi, N. Koji, M. Masami, K. Seiji
Effect of reduced pressure atmosphere on weld geometry in partial penetration laser welding of stainless steel and aluminium alloy with

high power and high brightness laser

Sci. Technol. Weld. Join., 19 (4) (2014), pp. 324-332, [10.1179/1362171813y.00000000182](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [15] Y. Arata, N. Abe, T. Oda
Fundamental phenomena in high power CO₂ laser (Report II): vacuum laser Welding(Welding Physics, Process & Instrument)
Trans. JWRI, 14 (2) (1985), pp. 217-222, [10.18910/11323](#) ↗
[Google Scholar](#) ↗
- [16] Y. Luo, X. Tang, F. Lu, Q. Chen, H. Cui
Effect of subatmospheric pressure on plasma plume in fiber laser welding
J. Mater. Process. Technol., 215 (2015), pp. 219-224, [10.1016/j.jmatprotec.2014.08.011](#) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [17] M. Jiang, N. Jiang, X. Chen, S. Ma, Y. Chen, Y. Chen, Z. Lei
Experimental and numerical investigation of single-pass laser welding of 20 mm-thick high-strength steel under reduced ambient pressure
J. Mater. Res. Technol. JmrT, 15 (2021), pp. 2317-2331, [10.1016/j.jmrt.2021.09.030](#) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [18] M. Bachmann, V. Avilov, A. Gumenyuk, M. Rethmeier
Numerical simulation of full-penetration laser beam welding of thick aluminium plates with inductive support
J. Phys. D. Appl. Phys., 45 (3) (2012), [10.1088/0022-3727/45/3/035201](#) ↗
[Google Scholar](#) ↗
- [19] V. Avilov, A. Fritzsche, M. Bachmann, A. Gumenyuk, M. Rethmeier
Full penetration laser beam welding of thick duplex steel plates with electromagnetic weld pool support
J. Laser Appl., 28 (2) (2016), [10.2351/1.4944103](#) ↗
[Google Scholar](#) ↗

- [20] L. Xu, X. Tang, S. Han, R. Zhang, C. Shao, H. Cui
Study on root hump formation mechanism in full penetration laser welding with backing gas for aluminum alloy
J. Mater. Process. Technol., 307 (2022), [10.1016/j.jmatprotec.2022.117674](https://doi.org/10.1016/j.jmatprotec.2022.117674) ↗
[Google Scholar](#) ↗
- [21] Y. Hu, X. Zhang, C. Wang, F. Yan, Z. Zhao, X. Yan
Suppression mechanism of collapse defects and microstructural evolution during full-penetration laser welding assisted by the following bottom airflow
J. Mater. Process. Technol., 332 (2024), [10.1016/j.jmatprotec.2024.118531](https://doi.org/10.1016/j.jmatprotec.2024.118531) ↗
[Google Scholar](#) ↗
- [22] D. Bergstrom, J. Powell, A.F.H. Kaplan
The absorptance of steels to Nd:YLF and Nd:YAG laser light at room temperature
Appl. Surf. Sci., 253 (11) (2007), pp. 5017-5028, [10.1016/j.apsusc.2006.11.018](https://doi.org/10.1016/j.apsusc.2006.11.018) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [23] R. Baumann, D. Hipp, A.F. Lasagni, A. Mahrle
Tailored absorptivity: increasing the laser weldability of copper through surface structuring
Mater. Lett., 283 (2021), [10.1016/j.matlet.2020.128700](https://doi.org/10.1016/j.matlet.2020.128700) ↗
[Google Scholar](#) ↗
- [24] M. Sokolov, A. Salminen
Experimental investigation of the influence of edge morphology in high power fiber laser welding
7th Conference on Laser Assisted Net shape Engineering (LANE) / International Conference on Photonic Technologies, Furth, GERMANY (2012), pp. 33-42
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [25] M. Sokolov, A. Salminen, Methods for improving laser beam welding efficiency, 8th International Conference on Laser Assisted Net Shape Engineering (LANE), Furth, GERMANY, 2014, pp. 450-457.

[Google Scholar ↗](#)

- [26] Y. Li, F. Wang, H. Liu, L. Yang, P. Zhang, Y. Wang, H. Li
Effect of surface roughness on the performances of laser-welded invar 36 alloy joints

Opt. Laser Technol., 162 (2023), [10.1016/j.optlastec.2023.109307 ↗](#)

[Google Scholar ↗](#)

- [27] C. Cai, Y. Liu, E. Wang, H. Chen, Z. Chen
Mechanisms of gap bridging and hump suppression in oscillating laser-arc hybrid welding of thick plate steel under variable gap conditions

J. Mater. Process. Technol., 339 (2025), [10.1016/j.jmatprotec.2025.118819 ↗](#)

[Google Scholar ↗](#)

- [28] S. Ozan, A. Bilgin, S. Kasman
Laser textured Ti-6Al-7Nb alloy for biomedical applications: an investigation of texturing parameters on surface properties

Proc. Inst. Mech. Eng. Part H. J. Eng. Med., 237 (10) (2023), pp. 1139-1153,

[10.1177/09544119231200537 ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [29] Q. Wang, Y. Cheng, Z. Zhu, N. Xiang, H. Wang
Modulation and control of wettability and hardness of Zr-Based metallic glass via facile laser surface texturing

Micromachines, 12 (11) (2021), [10.3390/mi12111322 ↗](#)

[Google Scholar ↗](#)

- [30] R.A. Alsaigh

Enhancement of surface properties using Ultrashort-Pulsed-Laser texturing: a review

Crystals, 14 (4) (2024), [10.3390/cryst14040353](#) ↗

[Google Scholar](#) ↗

- [31] J. Zhou, J. Wu, S. Tang, Y. Li
Review of laser texturing technology for surface protection and functional regulation of aluminum alloys: wettability, Anti-Icing, corrosion resistance, and wear resistance

Coatings, 15 (5) (2025), [10.3390/coatings15050567](#) ↗

[Google Scholar](#) ↗

- [32] F. Li, C. Li, J. Zhou, J. He, J. Wang, C. Luo, S. Li
Effect of laser parameters on surface texture of polyformaldehyde and parameter optimization

Stroj. Vestn. J. Mech. Eng., 70 (5-6) (2024), pp. 247-258, [10.5545/sv-jme.2023.787](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [33] Y. Su, L. Wang, Z. Yao, Q. Zhang, Z. Chen, J. Duan, T. Ye, J. Yao, - Hybrid Laser Cleaning of Carbon Deposits on N52B30 Engine Piston Crowns: Multi-Objective Optimization via Response Surface Methodology, - 18(- 15) (2025).

[Google Scholar](#) ↗

- [34] Z.-H. Lai, S. Xu, S.J. Clark, K. Fezzaa, J. Li
Unveiling mechanisms and onset threshold of humping in high-speed laser welding

Nat. Commun., 15 (1) (2024), [10.1038/s41467-024-53888-w](#) ↗

[Google Scholar](#) ↗

- [35] A.A. Lassila, E. Lundell, T. Andersson, D. Lonn, K. Salomonsson, R. Ghasemi

Experimental and numerical investigation of process-induced recoil force in keyhole laser welding: insights for validating multi-physics process simulations and modelling assumptions

J. Mater. Process. Technol., 341 (2025), [10.1016/j.jmatprotec.2025.118895](https://doi.org/10.1016/j.jmatprotec.2025.118895) ↗

[Google Scholar](#) ↗

- [36] J. Li, P. Jiang, S. Geng, J. Xiong
Numerical and experimental study on keyhole dynamics and pore formation mechanisms during adjustable-ring-mode laser welding of medium-thick aluminum alloy

Int. J. Heat. Mass Transf., 214 (2023), [10.1016/j.ijheatmasstransfer.2023.124443](https://doi.org/10.1016/j.ijheatmasstransfer.2023.124443) ↗

[Google Scholar](#) ↗

- [37] L.J. Zhang, J.X. Zhang, A. Gumenyuk, M. Rethmeier, S.J. Na
Numerical simulation of full penetration laser welding of thick steel plate with high power high brightness laser

J. Mater. Process. Technol., 214 (8) (2014), pp. 1710-1720, [10.1016/j.jmatprotec.2014.03.016](https://doi.org/10.1016/j.jmatprotec.2014.03.016) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [38] Y. Hu, C. Wang, Z. Liu, Z. Zhao, F. Yan
The formation mechanism and process regulation of collapse defects in high-power laser penetration welding of 20mm thick 316L stainless steel plates

Int. J. Heat. Mass Transf., 247 (2025), [10.1016/j.ijheatmasstransfer.2025.127123](https://doi.org/10.1016/j.ijheatmasstransfer.2025.127123) ↗

[Google Scholar](#) ↗

- [39] L. Guo, H. Wang, H. Liu, Y. Huang, Q. Wei, C.L.A. Leunge, Y. Wu, H. Wang
Quantifying the effects of gap on the molten pool and porosity formation in laser butt welding

Int. J. Heat. Mass Transf., 209 (2023), [10.1016/j.ijheatmasstransfer.2023.124143](https://doi.org/10.1016/j.ijheatmasstransfer.2023.124143) ↗

[Google Scholar](#) ↗

- [40] Y. Ai, J. Liu, C. Ye, S. Han
Analysis of the dynamic behaviors of molten pool and keyhole for the oscillating laser welding of dissimilar materials stake welded T-joints with a gap
Int. J. Heat. Mass Transf., 232 (2024), [10.1016/j.ijheatmasstransfer.2024.125877](https://doi.org/10.1016/j.ijheatmasstransfer.2024.125877) ↗
[Google Scholar](#) ↗
- [41] Y. Wang, P. Jiang, S. Geng, B. Xu, J. Zhao
Influence of the wrinkle surface structures on the vapor flow and keyhole stability in 20 kw high power laser welding
Int. J. Heat. Mass Transf., 193 (2022), [10.1016/j.ijheatmasstransfer.2022.122958](https://doi.org/10.1016/j.ijheatmasstransfer.2022.122958) ↗
[Google Scholar](#) ↗
- [42] W.-I. Cho, S.-J. Na, C. Thomy, F. Vollertsen
Numerical simulation of molten pool dynamics in high power disk laser welding
J. Mater. Process. Technol., 212 (1) (2012), pp. 262-275, [10.1016/j.jmatprotec.2011.09.011](https://doi.org/10.1016/j.jmatprotec.2011.09.011) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [43] Y. Zhao, X. Zhan, L. Ma, H. Bu, Y. Li, C. Ma
The energy synergistic mechanism of coaxial heterogeneous-wavelength hybrid laser beam and its effect on welding molten pool dynamics for aluminum alloy
J. Mater. Process. Technol., 333 (2024), [10.1016/j.jmatprotec.2024.118605](https://doi.org/10.1016/j.jmatprotec.2024.118605) ↗
[Google Scholar](#) ↗
- [44] S. Chen, Y. Zhao, S. Tian, Y. Gu, X. Zhan
Study on keyhole coupling and melt flow dynamic behaviors simulation of 2219 aluminum alloy T-joint during the dual laser beam bilateral synchronous welding
J. Manuf. Process., 60 (2020), pp. 200-212, [10.1016/j.jmapro.2020.10.063](https://doi.org/10.1016/j.jmapro.2020.10.063) ↗

[View PDF](#)[View article](#)[Google Scholar ↗](#)

- [45] X.H. Ye, X. Chen
Three-dimensional modelling of heat transfer and fluid flow in laser full-penetration welding
J. Phys. D. Appl. Phys., 35 (10) (2002), pp. 1049-1056, [10.1088/0022-3727/35/10/313 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [46] S. Chakraborty, P. Dutta
A generalized formulation for evaluation of latent heat functions in enthalpy-based macroscopic models for convection-diffusion phase change processes
Metall. Mater. Trans. BProcess. Metall. Mater. Process. Sci., 32 (3) (2001), pp. 562-564, [10.1007/s11663-001-0042-6 ↗](#)
[Google Scholar ↗](#)
- [47] H. Wang, Y. Shi, S. Gong
Effect of pressure gradient driven convection in the molten pool during the deep penetration laser welding
J. Mater. Process. Technol., 184 (1-3) (2007), pp. 386-392, [10.1016/j.jmatprotec.2006.12.015 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [48] R. Wang, Y. Lei, Y. Shi
Numerical simulation of transient temperature field during laser keyhole welding of 304 stainless steel sheet
Opt. Laser Technol., 43 (4) (2011), pp. 870-873, [10.1016/j.optlastec.2010.10.007 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [49] J.-H. Cho, S.-J. Na
Implementation of real-time multiple reflection and fresnel absorption of laser beam in keyhole
J. Phys. D. Appl. Phys., 39 (24) (2006), pp. 5372-5378, [10.1088/0022-3727/39/24/039 ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [50] S.-W. Han, J. Ahn, S.-J. Na
A study on ray tracing method for CFD simulations of laser keyhole welding: progressive search method
Weld. World, 60 (2) (2016), pp. 247-258, [10.1007/s40194-015-0289-1 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [51] G.Y. Soh, G.H. Yeoh, V. Timchenko
An algorithm to calculate interfacial area for multiphase mass transfer through the volume-of-fluid method
Int. J. Heat. Mass Transf., 100 (2016), pp. 573-581, [10.1016/j.ijheatmasstransfer.2016.05.006 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [52] X. Chen, H.X. Wang
A calculation model for the evaporation recoil pressure in laser material processing
J. Phys. D. Appl. Phys., 34 (17) (2001), pp. 2637-2642, [10.1088/0022-3727/34/17/310 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [53] Z. Gao, Y. Wu, J. Huang
Analysis of weld pool dynamic during stationary laser-MIG hybrid welding
Int. J. Adv. Manuf. Technol., 44 (9-10) (2009), pp. 870-879, [10.1007/s00170-008-1896-4 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [54] M.M. Pariona, A.F. Taques, L.A. Woiciechowski
The marangoni effect on microstructure properties and morphology of laser-treated Al-Fe alloy with single track by FEM: varying the laser beam velocity
Int. J. Heat. Mass Transf., 119 (2018), pp. 10-19, [10.1016/j.ijheatmasstransfer.2017.11.097 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [55] M. Gäumann, C. Bezençon, P. Canalis, W. Kurz
Single-crystal laser deposition of superalloys:: Processing-microstructure maps
Acta Mater., 49 (6) (2001), pp. 1051-1062, [10.1016/s1359-6454\(00\)00367-0](https://doi.org/10.1016/s1359-6454(00)00367-0) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [56] D. Liu, D. Wu, R. Wang, J. Shi, F. Niu, G. Ma
Formation mechanism of Al-Zn-Mg-Cu alloy fabricated by laser-arc hybrid additive manufacturing: microstructure evaluation and mechanical properties
Addit. Manuf., 50 (2022), [10.1016/j.addma.2021.102554](https://doi.org/10.1016/j.addma.2021.102554) ↗
[Google Scholar](#) ↗
- [57] C. Fang, J. Xin, W. Dai, J. Wei, J. Wu, Y. Song
Deep penetration laser welding of austenitic stainless steel thick-plates using a 20 kw fiber laser
J. Laser Appl., 32 (1) (2020), [10.2351/1.5094176](https://doi.org/10.2351/1.5094176) ↗
[Google Scholar](#) ↗
- [58] L. Gao, Y. Chen, X. Zhang, S.R. Agnew, A.C. Chuang, T. Sun
Evolution of dislocations during the rapid solidification in additive manufacturing
Nat. Commun., 16 (1) (2025), [10.1038/s41467-025-59988-5](https://doi.org/10.1038/s41467-025-59988-5) ↗
[Google Scholar](#) ↗
- [59] C. Liang, N. Wang, Y. Chen, C. Jiang, G. Wu, Q. Zhao, L. Zhu, J. Luo
Transition of low and high-angle grain boundaries during strain rate-induced dislocation storage and annihilation
Mater. Charact., 205 (2023), [10.1016/j.matchar.2023.113284](https://doi.org/10.1016/j.matchar.2023.113284) ↗
[Google Scholar](#) ↗
- [60] L. Wang, G. Zhang, J. Xu, Y. Li, Q. Chen, Y. Rong, Y. Huang

Effect of collapse and hump on thermomechanical behavior in high-power laser welding of 16-mm marine steel EH40

Int. J. Adv. Manuf. Technol., 120 (3-4) (2022), pp. 2003-2013, [10.1007/s00170-022-08872-3](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [61] S. Gao, Y. Xiao, J. Liu, R. Gholizadeh, M.-H. Park, S. Yoshida, W. Mao, Y. Tian, N. Tsuji
Polymer-like double necking deformation in an FeCoCrV medium entropy alloy

Mater. Res. Lett., 13 (7) (2025), pp. 709-718, [10.1080/21663831.2025.2504524](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [62] Z. Zheng, C. Shao, M. Liu, F. Lu
Formation and migration of bubbles under different laser oscillation paths during laser welding process of medium-thick al alloy

Int. J. Heat. Mass Transf., 239 (2025), [10.1016/j.ijheatmasstransfer.2024.126553](#) ↗

[Google Scholar](#) ↗

- [63] W. Ke, Y. Liu, F.B. Teshome, Z. Zeng
Numerical study on multiphase evolution and molten pool dynamics of underwater wet laser welding in shallow water environment

Int. J. Heat. Mass Transf., 220 (2024), [10.1016/j.ijheatmasstransfer.2023.124976](#) ↗

[Google Scholar](#) ↗

- [64] X. Jin, P. Berger, T. Graf
Multiple reflections and fresnel absorption in an actual 3D keyhole during deep penetration laser welding

J. Phys. D. Appl. Phys., 39 (21) (2006), pp. 4703-4712, [10.1088/0022-3727/39/21/030](#) ↗

[Google Scholar](#) ↗

- [65] Y.-X. Zhang, S.-W. Han, J. Cheon, S.-J. Na, X.-D. Gao

Effect of joint gap on bead formation in laser butt welding of stainless steel

J. Mater. Process. Technol., 249 (2017), pp. 274-284, [10.1016/j.jmatprotec.2017.05.040](https://doi.org/10.1016/j.jmatprotec.2017.05.040) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [66] I. Bunaziv, C. Dorum, X. Ren, M. Eriksson, O.M. Akselsen
Application of LBW and LAHW for fillet welds of 12 and 15 mm structural steel

17th Nordic Laser Materials Processing Conference (NOLAMP), Trondheim, NORWAY (2019), pp. 121-130



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [67] C. Ma, Y. Li, L. Cheng, Y. Zhao, J. Wang, X. Zhan
Numerical analysis of gravity-induced coupling dynamics of keyhole and molten pool in laser welding

Int. J. Therm. Sci., 201 (2024), [10.1016/j.ijthermalsci.2024.108987](https://doi.org/10.1016/j.ijthermalsci.2024.108987) ↗

[Google Scholar](#) ↗

- [68] J. Liu, F. Jiang, S. Tashiro, S. Chen, M. Tanaka, C. Zhang, B. Xu, G. Zhang
The mechanism of dual-stagnation points flow phenomenon in keyhole plasma arc melt pool and corresponding control strategy

J. Manuf. Process., 125 (2024), pp. 473-488, [10.1016/j.jmapro.2024.07.078](https://doi.org/10.1016/j.jmapro.2024.07.078) ↗



[View PDF](#) [View article](#) [Google Scholar](#) ↗

- [69] J. Yang, L.M. Schlenger, M.H. Nasab, S. Van Petegem, F. Marone, R.E. Loge, C. Leinenbach
Experimental quantification of inward marangoni convection and its impact on keyhole threshold in laser powder bed fusion of stainless steel

Addit. Manuf., 84 (2024), [10.1016/j.addma.2024.104092](https://doi.org/10.1016/j.addma.2024.104092) ↗

[Google Scholar](#) ↗

- [70] R. Zhang, X. Tang, L. Xu, F. Lu, H. Cui

Study of molten pool dynamics and porosity formation mechanism in full penetration fiber laser welding of Al-alloy

Int. J. Heat. Mass Transf., 148 (2020), [10.1016/j.ijheatmasstransfer.2019.119089](https://doi.org/10.1016/j.ijheatmasstransfer.2019.119089) ↗

[Google Scholar](#) ↗

[71]

J. Frostevarg

Factors affecting weld root morphology in laser keyhole welding

Opt. Lasers Eng., 101 (2018), pp. 89-98, [10.1016/j.optlaseng.2017.10.005](https://doi.org/10.1016/j.optlaseng.2017.10.005) ↗



[View PDF](#)

[View article](#)

[View in Scopus](#) ↗

[Google Scholar](#) ↗

[View Abstract](#)

Recommended articles

Based on reading popularity

Discussion Full text access

Comments on “Formation of $M_{23}C_6$ -type precipitates and chromium-depleted zones in austenite stainless steel, Scr. Mater. 65 (2011) 509–512”

Won Seok Choi, Hyunjoon Park

Scripta Materialia • Volume 209 • 2022 • Article 114412



[View PDF](#)

Research article Full text access

Phosphorous behavior and its effect on secondary phase formation in high refractory content powder-processed Ni-based superalloys

Stoichko Antonov, ..., Sammy Tin

Materialia • Volume 7 • 2019 • Article 100423



[View PDF](#)

Short communication Open access

Cu addition retards the static recrystallization of cold-deformed 316L biomedical stainless steel

Xiangfeng Du, ..., Ke Yang

Journal of Materials Research and Technology • Volume 19 • 2022 • Pages 1673-1677

 [View PDF](#)

Research article Full text access

Effect of the cellular structure on austenite distribution and cryogenic mechanical properties of maraging stainless steel processed by laser powder bed fusion

Chao Wang, ..., Weidong Huang

Materials Science and Engineering: A • Volume 947 • 2025 • Article 149243

 [View PDF](#)

Research article Full text access

Retained austenite stability in cementite-free bainitic steels processed by different isothermal heat treatments: Insights from quasi-in situ strain partition analysis

Xiedong Huang, ..., Quan Shan

Materials Characterization • Volume 229, Part A • 2025 • Article 115474

 [View PDF](#)

Research article Full text access

Stability of γ' precipitates revealed by in-situ heating and tension TEM in selective laser melted IN718Plus superalloy

Pinjie Wang, ..., Xin Lin

Materials Science and Engineering: A • Volume 944 • 2025 • Article 148917

 [View PDF](#)

Metrics

Captures

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.